TEST REPORT

IEC 61727 / IEC 62116

Photovoltaic (PV) systems
Characteristics of the utility interface
Test procedure of islanding prevention measures for utility-interconnected photovoltaic inverters

Report reference number .....:	**OMK-15OC1335FTSP**
Date of issue .....:.....	2015-11-03
Total number of pages .....:.....	72
Testing laboratory name .....:	**Bureau Veritas** **LCIE China Company Limited**
Address .....:	Building 4, No. 518, Xinzhuan Road, Caohejing Songjiang High-Tech Park, Shanghai, P.R. China (201612)
Applicant's name.....:	**Omnik New Energy Co., Ltd.**
Address .....:	Xinghu Road No.218 bioBAY Park A4-314, 215123 Suzhou, China
Test specification	
Standard.....:	IEC 61727:2004, EN 61727:1995, DIN EN 61727:1996 IEC 62116:2008, EN 62116:2011, DIN EN 62116:2012 IEC 62116:2014 With deviations for Thailand according to the grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA):2013
Certificate.....:	**Certificate of compliance**
Test report form number .....:	IEC 62116
Master TRF .....:	Bureau Veritas Consumer Products Services Germany GmbH
Test item description.....:	**Grid-tied photovoltaic inverter**
Trademark.....:	OMNIK
Model / Type .....:	Omniksol-5k-TL2

Ratings .....:	**Omniksol-5k-TL2**
MPP DC voltage range [V].....:	120-500
Input DC voltage range [V].....:	120-590
Input DC current [A] .....:	28(nom), 36(max)
Output AC voltage [V] .....:	230V/50Hz
Output AC current [A].....:	20(nom), 22(max)
Output power [VA].....:	4600(nom) 5000(max)

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Testing Location .....:	**BV LCIE China Company Limited**
Address .....:	Building 4, No. 518, Xinzhuan Road, Caohejing Songjiang High-Tech Park, Shanghai, P.R.China (201612)
Tested by (name and signature) .....:	Vic Zhou Test engineer Vic Zhou
Approved by (name and signature) .....:	Harvey Wang Senior project engineer Harvey Wang
Manufacturer's name .....:	**Omnik New Energy Co., Ltd.**
Factory address .....:	Xinghu Road No.218 bioBAY Park A4-314, 215123 Suzhou, China

Document History

Date	Internal reference	Modification / Change / Status	Revision
2015-11-03	Vic Zhou	Initial report was written	0

Supplementary information:

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Test items particulars

Equipment mobility ..... :	Permanent connection
Operating condition ..... :	Continuous
Class of equipment ..... :	Class I
Protection against ingress of water .. :	IP65 according to EN 60529
Mass of equipment [kg] ..... :	16,5 kg

Test case verdicts

Test case does not apply to the test object ..... :	N/A
Test item does meet the requirement ..... :	P(ass)
Test item does not meet the requirement ..... :	F(ail)

Testing

Date of receipt of test item ..... :	2015-10-27
Date(s) of performance test ..... :	2015-08-20 to 2015-09-20

General remarks:

The test result presented in this report relate only to the object(s) tested.
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”(see Annex #)" refers to additional information appended to the report.
"(see appended table)" refers to a table appended to the report.

Throughout this report a comma is used as the decimal separator.

The IEC61727 does not provide any limits of accuracy for the utility voltage and frequency measurement of the PV-system. If nothing different stated at the test table the values for tolerances given in EN 50438, Table 2 are used.

Tolerances on trip values tabel 2 EN50438:

- Voltage: +/- 1% of the nominal voltage
- Frequency: +/- 0,5% of the nominal frequency
- Clearance time: +/- 10%

This Test Report consists of the following documents:

1. Test Results
2. Annex No. 1 – EMC Test Report
3. Annex No. 2 – Pictures of the unit
4. Annex No. 3 – Test equipment list

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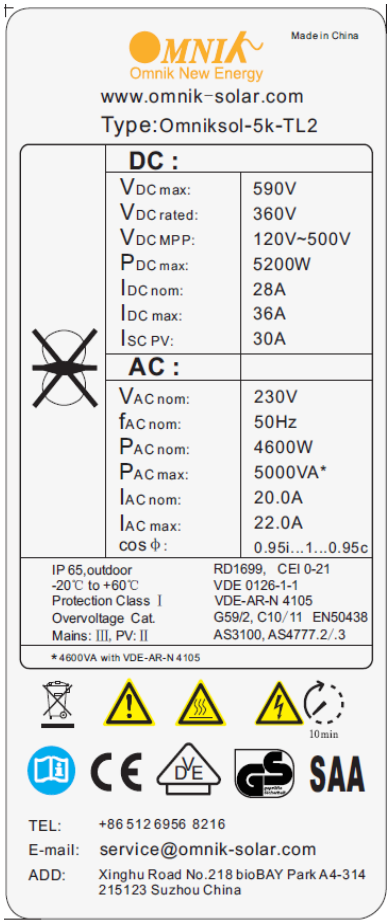
Copy of marking plate:



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General product information:

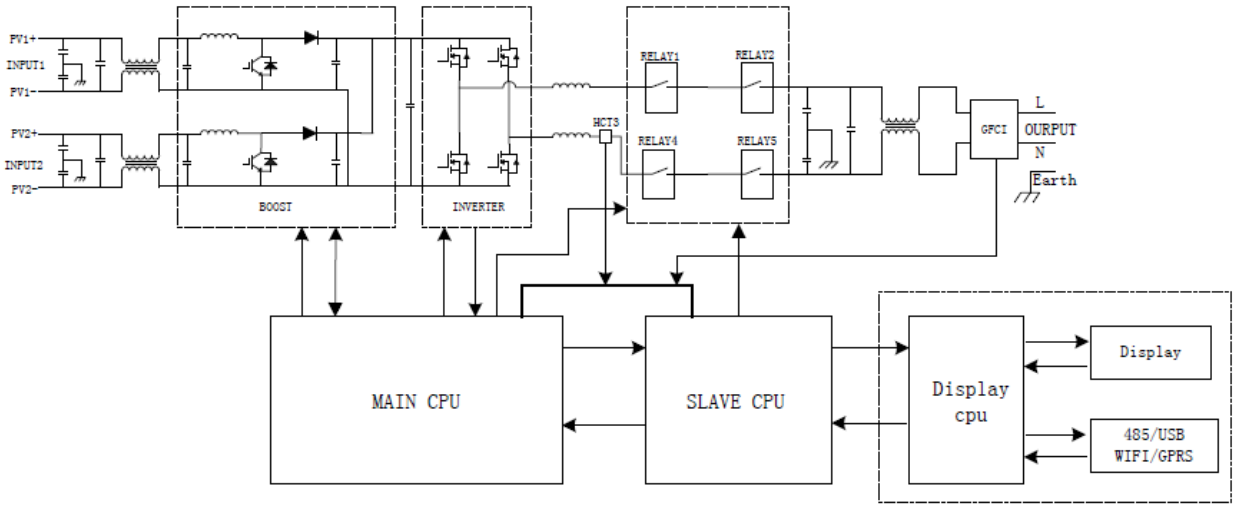
The Solar Inverter converts DC voltage into AC voltage.

The unit is providing EMC filtering at the output towards mains and at the input toward the solar array. The Solar converter is transformer-less type. The output is switched off redundant by the high power switching bridge and two relays. This assures that the opening of the output circuit will also operate in case of one error.

The internal control is redundant built. It consists out of two Microcontrollers CPU (U1, U5), The master CPU (U1) control the relays, measures grid voltage, frequency, PV current and voltage, DC bus voltage, AC current with injected DC and insulation resistance. In addition it tests the current sensors and the RMCU circuit before each start up
The redundant CPU (U5) also measures the voltage and current of grid side and PV side. It also shut down the relays. Both microcontrollers communicate with each other.
The voltage and frequency measurement is performed with resistors in serial which are connected directly to line and neutral. Both controllers get these signals and analyze the data.
The units provide two relays in series in each path (L and N). The relays are tested before each start up. In addition the power bridge can be stopped by both CPU.

Differences of the models:

Block diagrams: Omniksol-5k-TL2



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The product was tested on

Hardware version:

Model	Omniksol-5k-TL2
INVBST	999004-0-03-040
CTL	999004-0-01-031
HMI	999004-0-04-040
SPS	999004-0-02-012
COM	999004-0-05-020

Software Version:

Model	Omniksol-5k-TL2
Main CPU	V1.0Build 98
Slave CPU	V1.0Build 31
HMI CPU	V1.0Build 13

The maximum ambient temperature is specified as 60°C.

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Interface protection settings with deviations according to the grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA):2013 **(Thailand MEA)**		
Parameter	**Max. clearance time***	**Trip setting**
Over voltage (level 2)	0,05s	230V +35% (311V)
Over voltage (level 1)	2,0s	230V +4,3% (240V)
Under voltage (level 1)	2,0s	230V -13% (200V)
Under voltage (level 2)	0,1s	230V -50% (115V)
Over frequency	0,1s	50Hz +2% (51,0Hz)
Under frequency	0,1s	50Hz -2% (49,0Hz)
Reconnection time	at least 120s	
Permanent DC-injection	0,5% of rated inverter output current	
Loss of main IEC 62116	Inverter shall detect and disconnect within 0,3s	
* Trip time refers to the time between the abnormal condition occurring and the inverter ceasing to energize the utility line. The PV system control circuits shall actually remain connected to the utility to allow sensing of utility electrical conditions for use by the “reconnect” feature.		

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IEC61727:2004			
Clause	Requirement – Test	Result – Remark	Verdict
SECTION 4: Utility compatibility			
4	**General** The quality of power provided by the PV system for the on-site AC loads and for power delivered to the utility is governed by practices and standards on voltage, flicker, frequency, harmonics and power factor. Deviation from these standards represents out-of-bounds conditions and may require the PV system to sense the deviation and properly disconnect from the utility system. All power quality parameters (voltage, flicker, frequency, harmonics, and power factor) must be measured at the utility interface/ point of common coupling unless otherwise specified.	Noticed	**P**
4.1	**Voltage, current and frequency** The PV system AC voltage, current and frequency shall be compatible with the utility system.	Derived from tests	**P**
4.2	**Normal voltage operating range** Utility-interconnected PV systems do not normally regulate voltage; they inject current into the utility. Therefore, the voltage operating range for PV inverters is selected as a protection function that responds to abnormal utility conditions, not as a voltage regulation function.	Derived from tests	**P**
4.3	**Flicker** The operation of the PV system should not cause voltage flicker in excess of limits stated in the relevant sections of IEC 61000-3-3 for systems less than 16 A or IEC 61000-3-5 for systems with current of 16 A and above.	See table 4.3	**P**
4.4	**DC injection** The PV system shall not inject DC current greater than 1 % of the rated inverter output current, into the utility AC interface under any operating condition.	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See table 4.4	**P**
4.5	**Normal frequency operating range** The PV system shall operate in synchronism with the utility system, and within the frequency trip limits defined in 5.2.2.	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See table 4.5 and 5.2.2	**P**

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IEC61727:2004			
Clause	Requirement – Test	Result – Remark	Verdict
SECTION 4: Utility compatibility			
4.6	**Harmonics and waveform distortion** Low levels of current and voltage harmonics are desirable; the higher harmonic levels increase the potential for adverse effects on connected equipment. Acceptable levels of harmonic voltage and current depend upon distribution system characteristics, type of service, connected loads/apparatus, and established utility practice. The PV system output should have low current-distortion levels to ensure that no adverse effects are caused to other equipment connected to the utility system. Total harmonic current distortion shall be less than 5 % at rated inverter output. Each individual harmonic shall be limited to the percentages listed in Table 1. Even harmonics in these ranges shall be less than 25 % of the lower odd harmonic limits listed. (see Clause 4.6 Table 1 – Current distortion limits)	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See tables 4.6 (1) and 4.6 (2)	**P**
4.7	**Power factor** The PV system shall have a lagging power factor greater than 0,9 when the output is greater than 50 % of the rated inverter output power.	See table 4.7	**P**

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IEC61727:2004			
Clause	Requirement – Test	Result – Remark	Verdict
SECTION 5: Personnel safety and equipment protection			
5	**General** This Clause provides information and considerations for the safe and proper operation of the utility-connected PV systems.	Noticed	**P**
5.1	**Loss of utility voltage** To prevent islanding, a utility connected PV system shall cease to energize the utility system from a de-energized distribution line irrespective of connected loads or other generators within specified time limits. A utility distribution line can become de-energized for several reasons. For example, a substation breaker opening due to fault conditions or the distribution line switched out during maintenance. If inverters (single or multiple) have DC SELV input and have accumulated power below 1 kW then no mechanical disconnect (relay) is required.	The following deviations were used: a) Metropolitan Electricity Authority (MEA)	**P**
5.2	**Over/under voltage and frequency** Abnormal conditions can arise on the utility system that requires a response from the connected photovoltaic system. This response is to ensure the safety of utility maintenance personnel and the general public, as well as to avoid damage to connected equipment, including the photovoltaic system. The abnormal utility conditions of concern are voltage and frequency excursions above or below the values stated in this Clause, and the complete disconnection of the utility, presenting the potential for a distributed resource island.	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See table 5.2.1 and 5.2.2	**P**
5.2.1	**Over/under voltage** When the interface voltage deviates outside the conditions specified in Table 2, the photovoltaic system shall cease to energize the utility distribution system. This applies to any phase of a multiphase system. All discussions regarding system voltage refer to the local nominal voltage. The system shall sense abnormal voltage and respond. The following conditions should be met, with voltages in RMS and measured at the point of utility connection. (see clause 5.2.1 Table 2 – Response to abnormal voltages) The purpose of the allowed time delay is to ride through short-term disturbances to avoid excessive nuisance tripping. The unit does not have to cease to energize if the voltage returns to the normal utility continuous operation condition within the specified trip time.	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See table 5.2.1	**P**

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IEC61727:2004			
Clause	Requirement – Test	Result – Remark	Verdict
SECTION 5: Personnel safety and equipment protection			
5.2.2	**Over/under frequency** When the utility frequency deviates outside the specified conditions the photovoltaic system shall cease to energize the utility line. The unit does not have to cease to energize if the frequency returns to the normal utility continuous operation condition within the specified trip time. When the utility frequency is outside the range of ±1 Hz, the system shall cease to energize the utility line within 0,2 s. The purpose of the allowed range and time delay is to allow continued operation for short-term disturbances and to avoid excessive nuisance tripping in weak-utility system conditions.	The following deviations were used: a) Metropolitan Electricity Authority (MEA) See table 5.2.2	**P**
5.3	**Islanding protection** The PV system must cease to energize the utility line within 2 s of loss of utility.	The following deviations were used: a) Metropolitan Electricity Authority (MEA)	**P**
5.4	**Response to utility recovery** Following an out-of-range utility condition that has caused the photovoltaic system to cease energizing, the photovoltaic system shall not energize the utility line for 20 s to 5 min after the utility service voltage and frequency have recovered to within the specified ranges**.**	See table 5.2 (1) and 5.2 (2)	**P**
5.5	**Earthing** The utility interface equipment shall be earthed/grounded in accordance with IEC 60364-7-712.	Stated in the manual.	**P**
5.6	**Short circuit protection** The photovoltaic system shall have short-circuit protection in accordance with IEC 60364-7-712.	Stated in the manual.	**P**
5.7	**Isolation and switching** A method of isolation and switching shall be provided in accordance with IEC 60364-7-712.	Stated in the manual.	**P**

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Test overview:		
IEC 61727:2004		
Clause	**Test**	**Result**
1	Response to protection operation - fault condition tests (according VDE0126-1-1:2006)	**P**
4	Type test:	
4.3	Voltage Fluctuations and Flicker	**P**
4.4	Monitoring of DC-Injection	**P**
4.5	Normal frequency operating range (see 5.2.2 below)	**P**
4.6	Harmonics and waveform distortion	**P**
4.7	Power factor	**P**
5.2.1	Voltage monitoring	**P**
5.2.2	Frequency monitoring	**P**

IEC 62116:2008		
Clause	**Test**	**Result**
	Type test:	
6.1	Islanding protection according table 6 - Load imbalance (real, reactive load) for test condition A (EUT ouput = 100%)	**P**
6.1	Load imbalance (reactive load) for test condition B (EUT output = 50 % – 66 %)	**P**
6.1	Load imbalance (reactive load) for test condition C (EUT output = 25 % – 33 %)	**P**

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Test Results

1. Response to protection operation - fault condition tests			P
	ambient temperature [°C] :	24	—
	model/type of power supply :	AC: type 61512 DC: type 62150H-1000S	—
	manufacturer of power supply :	AC: Chroma DC: Chroma	—
	rated markings of power supply :	AC: 18kW three phase DC: 15kW, 15A, 1000V	—

Model: Omniksol-5k-TL2

component No.	fault	test condition		test time	fuse No, (AC)	fault condition		result
		AC	DC			AC	DC	
Relay RYA1	short circuit before start up	230	360	10Min	FA1	<0,1	<0,1	The inverter can't start up, error message: Fault 06. No damage, no hazard.
Relay RYA2	short circuit before start up	230	360	10Min	FA1	<0,1	<0,1	The inverter can't start up, error message: Fault 07. No damage, no hazard.
Relay RYA4	short circuit before start up	230	360	10Min	FA1	<0,1	<0,1	The inverter can't start up, error message: Fault 06. No damage, no hazard.
Relay RYA5	short circuit before start up	230	360	10Min	FA1	<0,1	<0,1	The inverter can't start up, error message: Fault 07. No damage, no hazard.
Output	Short circuit	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately, error message: Fault 8. The inverter did not feed in power to the grid. No damage, no hazard.
DC input	Reversed	230	360	10Min	FA1	<0,1	<0,1	The inverter did not feed in power to the grid. No damage, no hazard.
Output	Over load	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately, error message: Fault 18. The inverter did not feed in power to the grid. No damage, no hazard.
TX3	Short circuit pin(21) to pin (19)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. Warning LED on red. The inverter did not feed in power to the grid. No damage, no hazard.
TX3	Short circuit pin(18) to pin (19)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. Warning LED on red. The inverter did not feed in power to the grid. No damage, no hazard.

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component No.	fault	test condition		test time	fuse No, (AC)	fault condition		result
		AC	DC			AC	DC	
UA1	Short circuit pin(5) to pin (6)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. QA6 damaged. The inverter did not feed in power to the grid. No damage, no hazard.
UA7	Short circuit pin(5) to pin (6)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. QA20 damaged. The inverter did not feed in power to the grid, no hazard.
CA11	Short circuit	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
TX3	Short circuit pin(1) to pin (2)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. D39,Tx3 damaged. The inverter did not feed in power to the grid. No damage, no hazard.
TX3	Short circuit pin(4) to pin (5)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. D36,Tx3 damaged. The inverter did not feed in power to the grid. No damage, no hazard.
QB2	Short circuit pin(1) to pin (2)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
QB2	Short circuit pin(1) to pin (3)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
QA8	Short circuit pin(2) to pin (3)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
QA5	Short circuit pin(2) to pin (3)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
UB1	Short circuit pin(6) to pin (8)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. The inverter did not feed in power to the grid. No damage, no hazard.
UB4	Short circuit pin(1) to pin (2)	230	360	10Min	FA1	15,2	10,3	Unit operated normally, no damage, no hazard
UB4	Short circuit pin(3) to pin (4)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. Warning LED on red. The inverter did not feed in power to the grid. No damage, no hazard.
UB2	Short circuit pin(1) to pin (2)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. QA4,QA8,QA5 damaged. The inverter did not feed in power to the grid, no hazard.

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component No.	fault	test condition		test time	fuse No, (AC)	fault condition		result
		AC	DC			AC	DC	
UB2	Short circuit pin(3) to pin (4)	230	360	10Min	FA1	<0,1	<0,1	PCE shutdown immediately. QA4,QA8,damaged. The inverter did not feed in power to the grid, no hazard.
Note: **The errors in the control circuit simulate that the safety is even under one error ensured.**								

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4.3 Voltage fluctuation and flicker			P
Inverter >16A			
Limit	dc% = 3,3	P_{st}=1,0	P_{lt}=0,65*
Test value	See below		

Omniksol-5k-TL2

Omniksol-5k-TL2

AnalysisDate(MeasureDate) : Mon Mar 03 17:28:13 2014 (Mon Mar 03 17:27:13 2014)
Comment : Omnik

Regulation : IEC61000-3-11 Ed1.0
Interval : 10Min0Sec
Model : YOKOGAWA WT3000
Wiring : single-phase 2wire
Voltage Range : 300.00V
Voltage U1 : 230.44V
Set Frequency : 50Hz
Frequency U1 : 50.002Hz
Element : 1

PASS

Compatibility Condition : Compliance with IEC61000-3-3
Element : Pass
Total Element : Pass
dc (3.30%) : Pass
dmax (4.00%) : Pass
d(t) (500ms) : Pass
Pst (1.00%) : Pass
Plt (0.65) : Pass

No.	dc[%]	dmax[%]	d(t)[ms]	Pst
1	0.81 Pass	0.90 Pass	0.00 Pass	0.13 Pass
2	0.80 Pass	0.97 Pass	0.00 Pass	0.19 Pass
3	0.82 Pass	0.88 Pass	0.00 Pass	0.19 Pass
4	0.83 Pass	0.92 Pass	0.00 Pass	0.13 Pass
5	0.81 Pass	0.83 Pass	0.00 Pass	0.13 Pass
6	0.82 Pass	0.92 Pass	0.00 Pass	0.19 Pass
7	0.83 Pass	0.92 Pass	0.00 Pass	0.19 Pass
8	0.82 Pass	0.85 Pass	0.00 Pass	0.19 Pass
9	0.77 Pass	0.85 Pass	0.00 Pass	0.13 Pass
10	0.83 Pass	0.87 Pass	0.00 Pass	0.13 Pass
11	0.81 Pass	0.87 Pass	0.00 Pass	0.19 Pass
12	0.82 Pass	0.90 Pass	0.00 Pass	0.19 Pass
				Plt
				0.17 Pass

Note:
Mains Impedance according EN61000-3-11: **R_{max} = 0,24Ω; jX_{max}= 0,15Ω @50Hz (|Z_{max}| = 0,283Ω)**

Calculation of the maximum permissible grid impedance at the point of common coupling based on d_c:
$Z_{max} = Z_{ref} * 3,3\% / d_c(P_n)$

The tests should be based on the limits of the EN61000-3-11 for more than 16A.

* Technical requiremnts for Connecting Small Scale PV (ssPV) Systems to Low Voltage Distribution Networks: Limit Pst=1,0 and Plt=0,8

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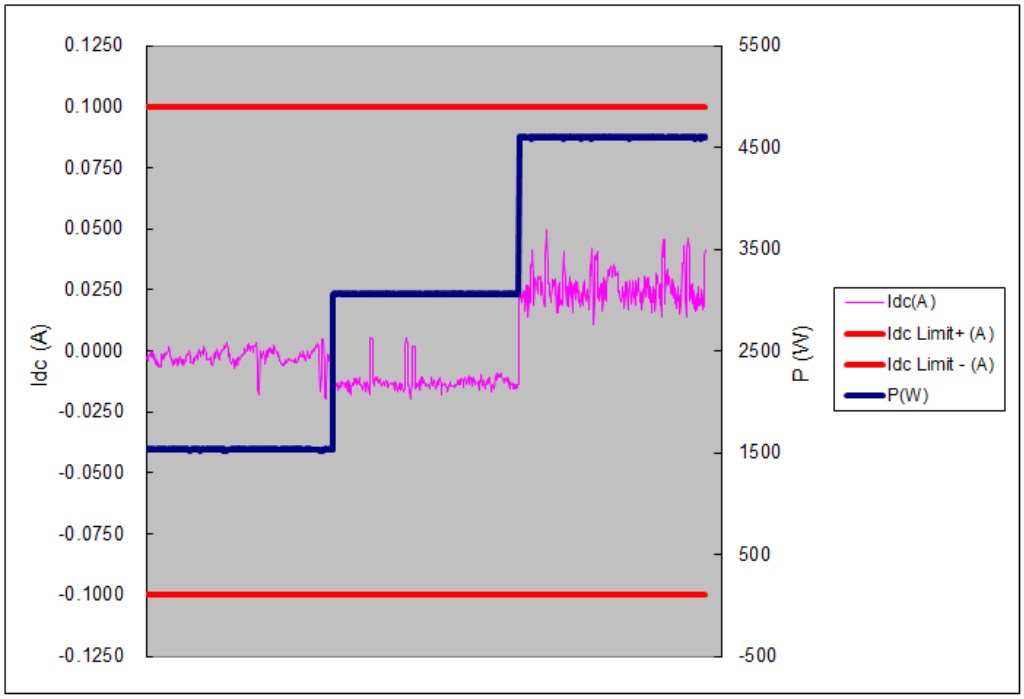
4.4 Monitoring of Permanent DC-Injection The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)			P
Omniksol-5k-TL2			
MEA and PEA Limit:	**0,5% of I_{nom} (100mA)**		
Output power:	33%	66%	100%
mean test value phase L1 [mA]:	-19,6	-19,9	49,8
mean test value phase L2 [mA]:	N/A	N/A	N/A
mean test value phase L3 [mA]:	N/A	N/A	N/A

Diagram of permanent DC injection: Omniksol-5k-TL2



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4.6 Harmonic Current Limit Test The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)				P
100% Output Power				
Watts [W]		**4545,8**		
Vrms [V]		**230,0**		
Arms [A]		**19,77**		
PF [1]		**0,9997**		
Frequency [Hz]		**50**		
THD50 [%]		**1,15**		
Harmonics	**Current Magnitude [A]**	**% of Fundamental**	**Phase**	**Harmonic Current Limits [%]**
1st	19,763	--	Single Phase	
2nd	0,016	0,082	Single Phase	1%
3rd	0,020	0,101	Single Phase	4%
4th	0,002	0,012	Single Phase	1%
5th	0,212	1,070	Single Phase	4%
6th	0,004	0,020	Single Phase	1%
7th	0,024	0,123	Single Phase	4%
8th	0,012	0,063	Single Phase	1%
9th	0,012	0,059	Single Phase	4%
10th	0,009	0,046	Single Phase	1%
11th	0,027	0,135	Single Phase	2%
12th	0,012	0,062	Single Phase	0,5%
13th	0,011	0,058	Single Phase	2%
14th	0,005	0,027	Single Phase	0,5%
15th	0,016	0,082	Single Phase	2%
16th	0,003	0,016	Single Phase	0,5%
17th	0,015	0,076	Single Phase	1,5%
18th	0,008	0,041	Single Phase	0,375%
19th	0,009	0,045	Single Phase	1,5%
20th	0,004	0,020	Single Phase	0,375%
21th	0,016	0,079	Single Phase	1,5%
22th	0,012	0,063	Single Phase	0,375%
23th	0,013	0,064	Single Phase	0,6%
24th	0,012	0,060	Single Phase	0,15%
25th	0,011	0,055	Single Phase	0,6%
26th	0,007	0,037	Single Phase	0,15%
27th	0,007	0,036	Single Phase	0,6%
28th	0,007	0,037	Single Phase	0,15%
29th	0,011	0,058	Single Phase	0,6%
30th	0,007	0,037	Single Phase	0,15%
31th	0,002	0,012	Single Phase	0,6%
32th	0,008	0,040	Single Phase	0,15%
33th	0,016	0,079	Single Phase	0,6%
34th	0,007	0,036	Single Phase	0,15%
35th	0,016	0,082	Single Phase	0,3%
36th	0,008	0,039	Single Phase	0,075%
37th	0,018	0,092	Single Phase	0,3%
38th	0,010	0,049	Single Phase	0,075%
39th	0,011	0,053	Single Phase	0,3%
40th	0,009	0,048	Single Phase	0,075%

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4.6 Harmonic Current Limit Test The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)				P
66% Output Power				
Watts [W]			**3000,0**	
Vrms [V]			**229,1**	
Arms [A]			**13,11**	
PF [1]			**0,9996**	
Frequency [Hz]			**50**	
THD50 [%]			**1,02**	
Harmonics	**Current Magnitude [A]**	**% of Fundamental**	**Phase**	**Harmonic Current Limits [%]**
1st	13,088	--	Single Phase	
2nd	0,012	0,090	Single Phase	1%
3rd	0,061	0,467	Single Phase	4%
4th	0,008	0,058	Single Phase	1%
5th	0,095	0,727	Single Phase	4%
6th	0,006	0,048	Single Phase	1%
7th	0,030	0,225	Single Phase	4%
8th	0,007	0,057	Single Phase	1%
9th	0,009	0,069	Single Phase	4%
10th	0,005	0,041	Single Phase	1%
11th	0,019	0,142	Single Phase	2%
12th	0,009	0,065	Single Phase	0,5%
13th	0,013	0,099	Single Phase	2%
14th	0,004	0,028	Single Phase	0,5%
15th	0,014	0,110	Single Phase	2%
16th	0,008	0,060	Single Phase	0,5%
17th	0,019	0,144	Single Phase	1,5%
18th	0,008	0,063	Single Phase	0,375%
19th	0,017	0,128	Single Phase	1,5%
20th	0,006	0,048	Single Phase	0,375%
21th	0,021	0,158	Single Phase	1,5%
22th	0,012	0,094	Single Phase	0,375%
23th	0,016	0,125	Single Phase	0,6%
24th	0,006	0,045	Single Phase	0,15%
25th	0,013	0,100	Single Phase	0,6%
26th	0,007	0,050	Single Phase	0,15%
27th	0,005	0,038	Single Phase	0,6%
28th	0,004	0,030	Single Phase	0,15%
29th	0,011	0,085	Single Phase	0,6%
30th	0,005	0,040	Single Phase	0,15%
31th	0,007	0,057	Single Phase	0,6%
32th	0,005	0,038	Single Phase	0,15%
33th	0,010	0,077	Single Phase	0,6%
34th	0,007	0,056	Single Phase	0,15%
35th	0,012	0,089	Single Phase	0,3%
36th	0,004	0,030	Single Phase	0,075%
37th	0,009	0,068	Single Phase	0,3%
38th	0,007	0,054	Single Phase	0,075%
39th	0,007	0,057	Single Phase	0,3%
40th	0,005	0,041	Single Phase	0,075%

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4.6 Harmonic Current Limit Test The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)				P
33% Output Power				
Watts [W]		**1528,8**		
Vrms [V]		**228,1**		
Arms [A]		**6,71**		
PF [1]		**0,9991**		
Frequency [Hz]		**50**		
THD50 [%]		**2,45**		
Harmonics	**Current Magnitude [A]**	**% of Fundamental**	**Phase**	**Harmonic Current Limits [%]**
1st	6,705	--	Single Phase	--
2nd	0,011	0,159	Single Phase	1%
3rd	0,102	1,515	Single Phase	4%
4th	0,011	0,163	Single Phase	1%
5th	0,052	0,776	Single Phase	4%
6th	0,013	0,197	Single Phase	1%
7th	0,027	0,409	Single Phase	4%
8th	0,006	0,084	Single Phase	1%
9th	0,031	0,458	Single Phase	4%
10th	0,013	0,194	Single Phase	1%
11th	0,056	0,828	Single Phase	2%
12th	0,004	0,063	Single Phase	0,5%
13th	0,023	0,338	Single Phase	2%
14th	0,006	0,089	Single Phase	0,5%
15th	0,031	0,455	Single Phase	2%
16th	0,006	0,087	Single Phase	0,5%
17th	0,026	0,381	Single Phase	1,5%
18th	0,007	0,103	Single Phase	0,375%
19th	0,031	0,465	Single Phase	1,5%
20th	0,012	0,174	Single Phase	0,375%
21th	0,013	0,190	Single Phase	1,5%
22th	0,003	0,039	Single Phase	0,375%
23th	0,016	0,231	Single Phase	0,6%
24th	0,010	0,142	Single Phase	0,15%
25th	0,020	0,302	Single Phase	0,6%
26th	0,007	0,108	Single Phase	0,15%
27th	0,010	0,152	Single Phase	0,6%
28th	0,006	0,085	Single Phase	0,15%
29th	0,009	0,130	Single Phase	0,6%
30th	0,003	0,042	Single Phase	0,15%
31th	0,010	0,154	Single Phase	0,6%
32th	0,003	0,044	Single Phase	0,15%
33th	0,013	0,197	Single Phase	0,6%
34th	0,006	0,089	Single Phase	0,15%
35th	0,009	0,136	Single Phase	0,3%
36th	0,003	0,041	Single Phase	0,075%
37th	0,009	0,141	Single Phase	0,3%
38th	0,002	0,036	Single Phase	0,075%
39th	0,006	0,092	Single Phase	0,3%
40th	0,002	0,032	Single Phase	0,075%

Note:
The harmonics are tested and evaluated according the IEEE1547.1-2005 clause 5.11.1 according the grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)

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4.7 Power factor **The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)**					**P**
	Omniksol-5k-TL2				
Output power [kW] **Test AC voltage [V]**	**~10%** **0,45kW**	**~25%** **1,15kW**	**~50%** **2,31kW**	**~75%** **3,43kW**	**~100%** **4,56kW**
230V	0,9829c	0,9979c	0,9993i	0,9995i	0,9996i

Note:
The PV system shall have a lagging power factor greater than 0,95 when the output is greater than 50% of the rated inverter output power.

The letter “i” is short for “inductive” and indicates inductive power factor. In case of capacitive power factor the letter “c” is used instead.

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5.2.1 Voltage monitoring **The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)**										**P**
First Level										
	Under Voltage					**Over Voltage**				
		Voltage [V]					Voltage [V]			
Set value		**199V**					**241V**			
Measured trip value		All	L1	L2	L3		All	L1	L2	L3
		N/A	199,1	N/A	N/A		N/A	240,9	N/A	N/A
		N/A	199,2	N/A	N/A		N/A	240,8	N/A	N/A
		N/A	199,1	N/A	N/A		N/A	240,8	N/A	N/A
		N/A	199,1	N/A	N/A		N/A	240,8	N/A	N/A
		N/A	199,2	N/A	N/A		N/A	240,9	N/A	N/A
Parameter		Time [ms]					Time [ms]			
Limit		**<= 2,0s**					**<= 2,0s**			
Trip time	204V to 194V	All	L1	L2	L3	236V to 246V	All	L1	L2	L3
		N/A	1,832	N/A	N/A		N/A	1,828	N/A	N/A
		N/A	1,840	N/A	N/A		N/A	1,824	N/A	N/A
		N/A	1,824	N/A	N/A		N/A	1,820	N/A	N/A
		N/A	1,840	N/A	N/A		N/A	1,820	N/A	N/A
		N/A	1,832	N/A	N/A		N/A	1,820	N/A	N/A
Reconnection time	at least 120s	137,0s				at least 120s	137,2s			

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Second Level										
	Under Voltage					**Over Voltage**				
Parameter		Voltage [V]					Voltage [V]			
Set value		**114V**					**272V***			
Measured trip value		All	L1	L2	L3		All	L1	L2	L3
		N/A	114,5	N/A	N/A		N/A	272,4	N/A	N/A
		N/A	114,6	N/A	N/A		N/A	272,4	N/A	N/A
		N/A	114,5	N/A	N/A		N/A	272,4	N/A	N/A
		N/A	114,5	N/A	N/A		N/A	272,4	N/A	N/A
		N/A	114,6	N/A	N/A		N/A	272,4	N/A	N/A
Parameter		Time [ms]					Time [ms]			
Limit		**<= 100ms**					**<= 50ms**			
Trip time	119V to 109V**	All	L1	L2	L3	267V to 277V**	All	L1	L2	L3
		N/A	73,0	N/A	N/A		N/A	21,0	N/A	N/A
		N/A	81,0	N/A	N/A		N/A	21,0	N/A	N/A
		N/A	73,0	N/A	N/A		N/A	21,0	N/A	N/A
		N/A	81,0	N/A	N/A		N/A	19,0	N/A	N/A
		N/A	65,0	N/A	N/A		N/A	19,0	N/A	N/A
Reconnection time	at least 120s	137,0s				at least 120s	137,0s			

Note:
*The max trip voltage is 272V for Omniksol-5k-TL2 set by manufactory
**The second trip level procedure(for under voltage) is a double jump, first set the voltage to 204V, and then within the disconnection time of the first level a jump to 119V and keep 500ms, second a jump to 109V to measure the disconnection time of the second voltage trips from 119V to 109V.

**The second trip level procedure(for over voltage) is a double jump, first set the voltage to 236V, and then within the disconnection time of the first level a jump to 267V and keep 500ms, second a jump to 277V to measure the disconnection time of the second voltage trips from 267V to 277V.

The voltage monitoring was tested according the test procedure 4.3.4 Operating voltage range test of the grid-connected inverter regulation Metropolitan Electricity Authority 2013. The voltage settings of the EUT are set for the tests as stated to 199V, 199V for undervoltage and 241V, 272V for overvoltage. The tests are performed according the IEEE 1547.1-2005, annex A. The reconnection time follows the test procedure of IEEE 1547.1, clause 5.10.

If inverter cannot be adjusted for overvoltage trip setting and/or undervoltage trip setting up to 311 V and/or 114 V respectively, it shall be adjusted the overvoltage trip setting and/or undervoltage trip setting to the maximum voltage and/or minimum voltage that inverter can be set.

The voltage which inverter stops feeding power to electrical system in each test must be in the range of the Voltage Trip Setting ± 1 V and the time it takes to cut off the power must be as specified in clause 3.2.1.

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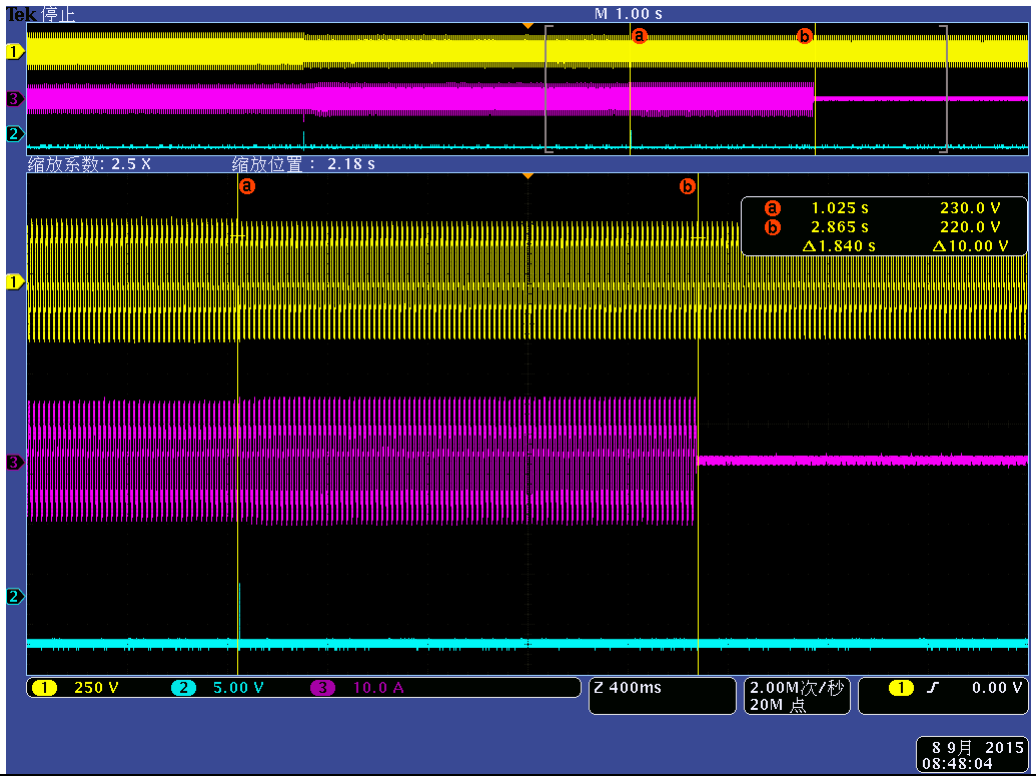
Under Voltage First Level single phase



Over Voltage First Level single phase

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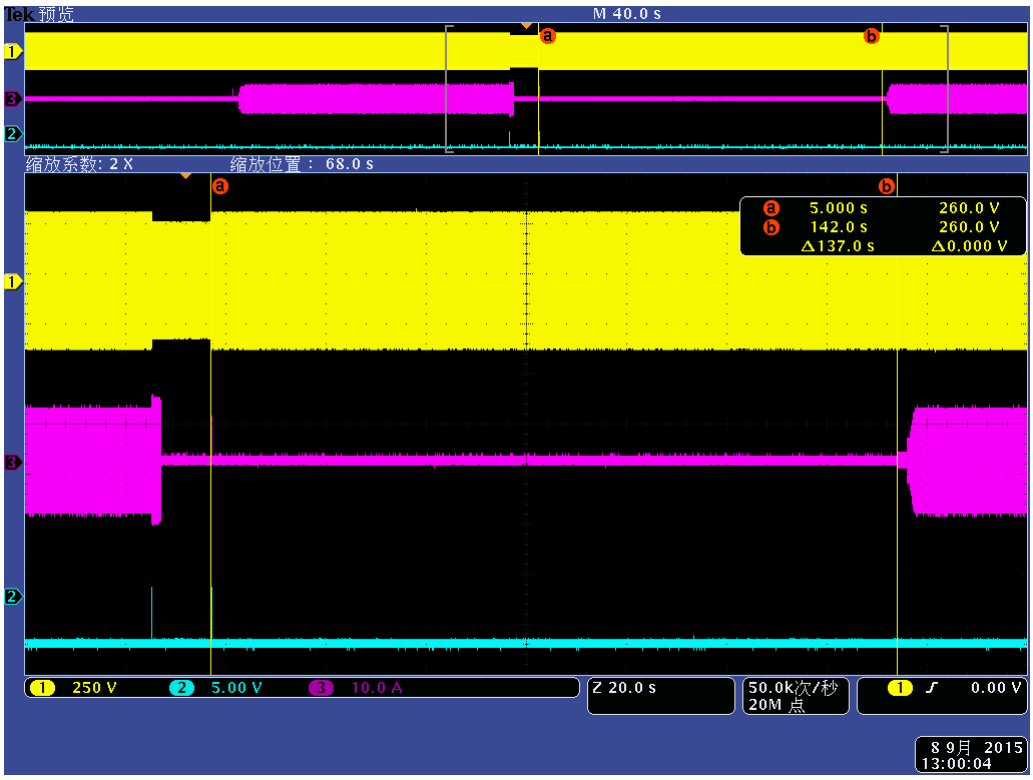
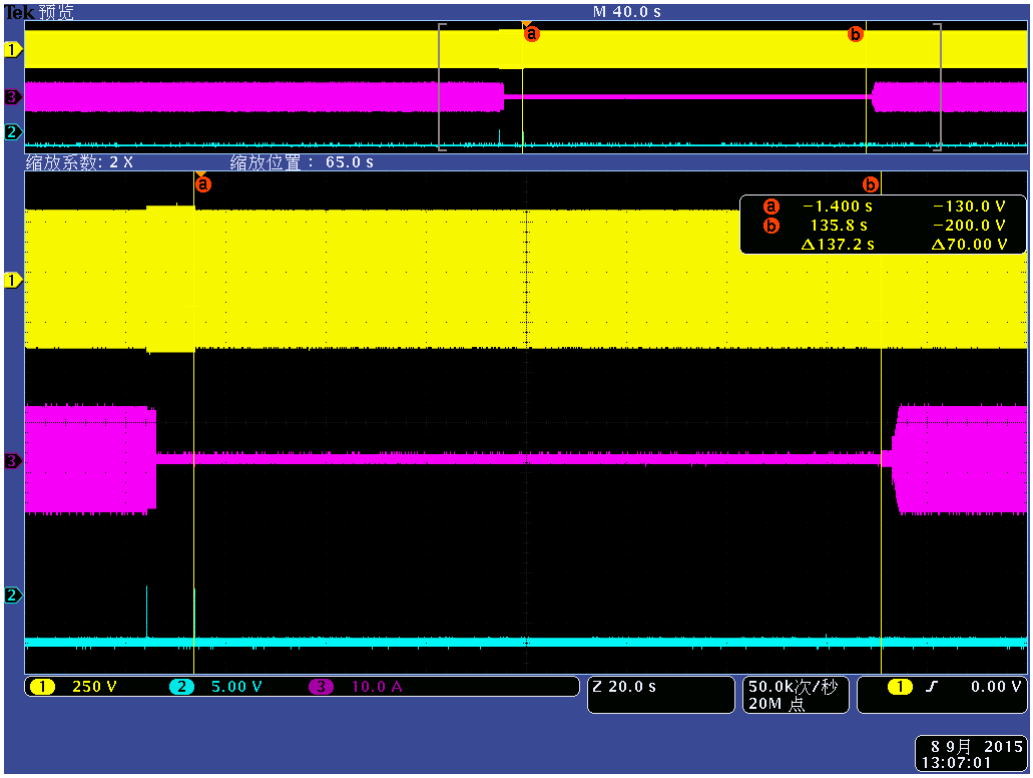
Reconnection Under Voltage first Level



Reconnection Over Voltage first Level



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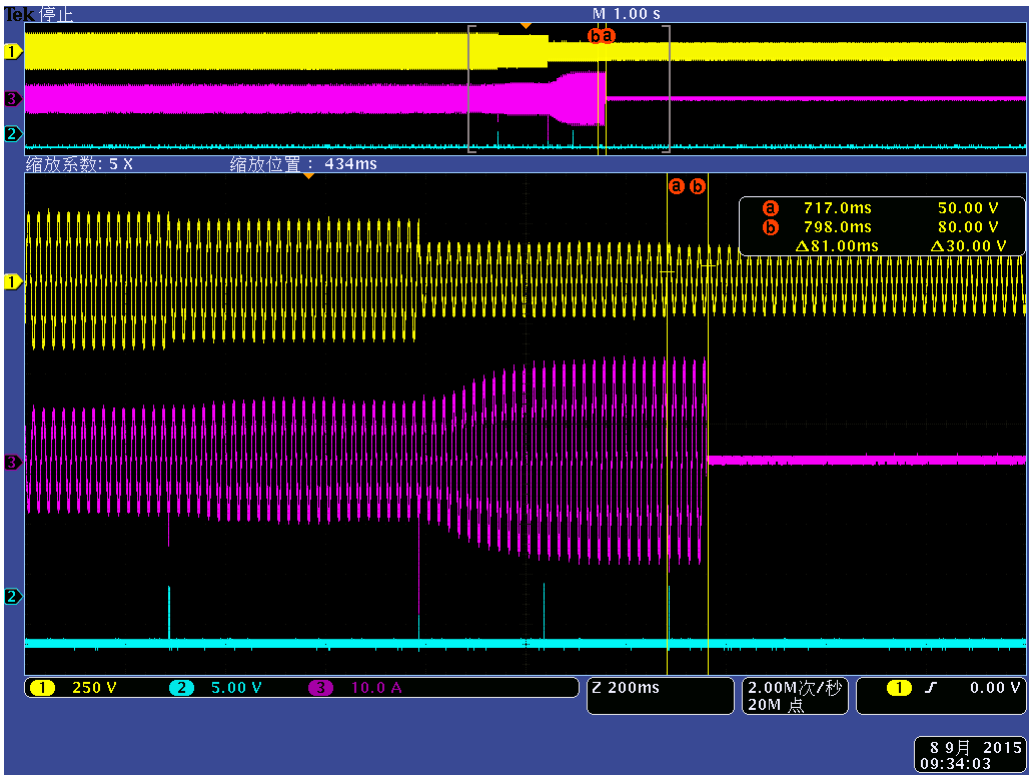
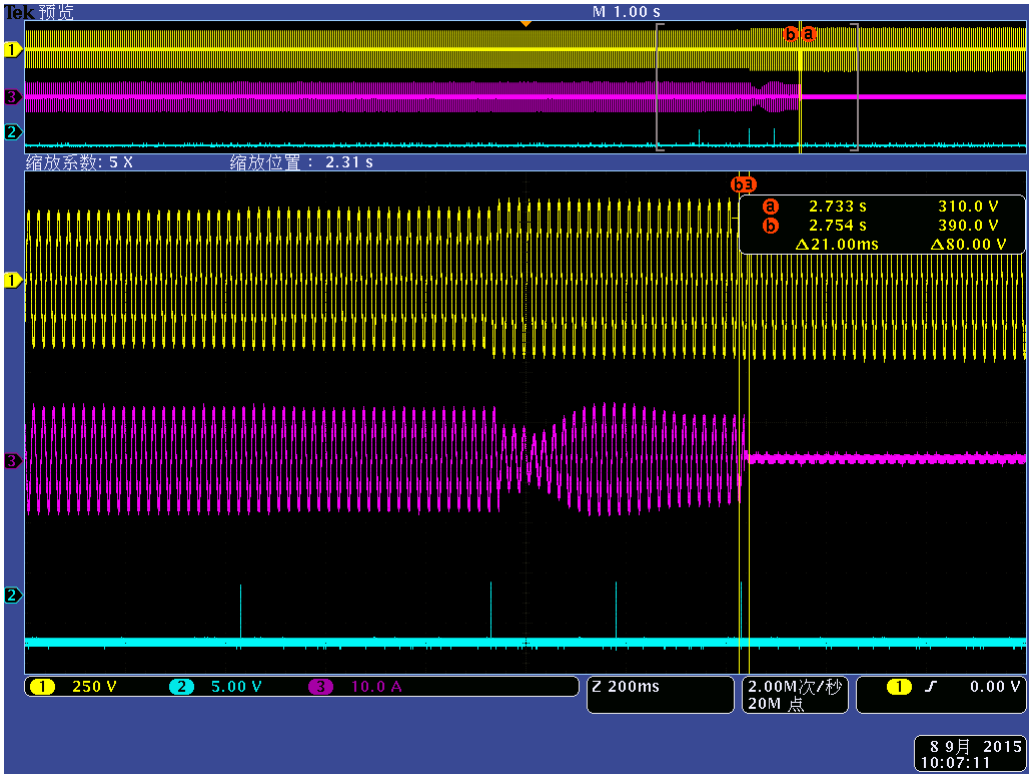
Under Voltage Second Level single phase



Over Voltage Second Level single phase



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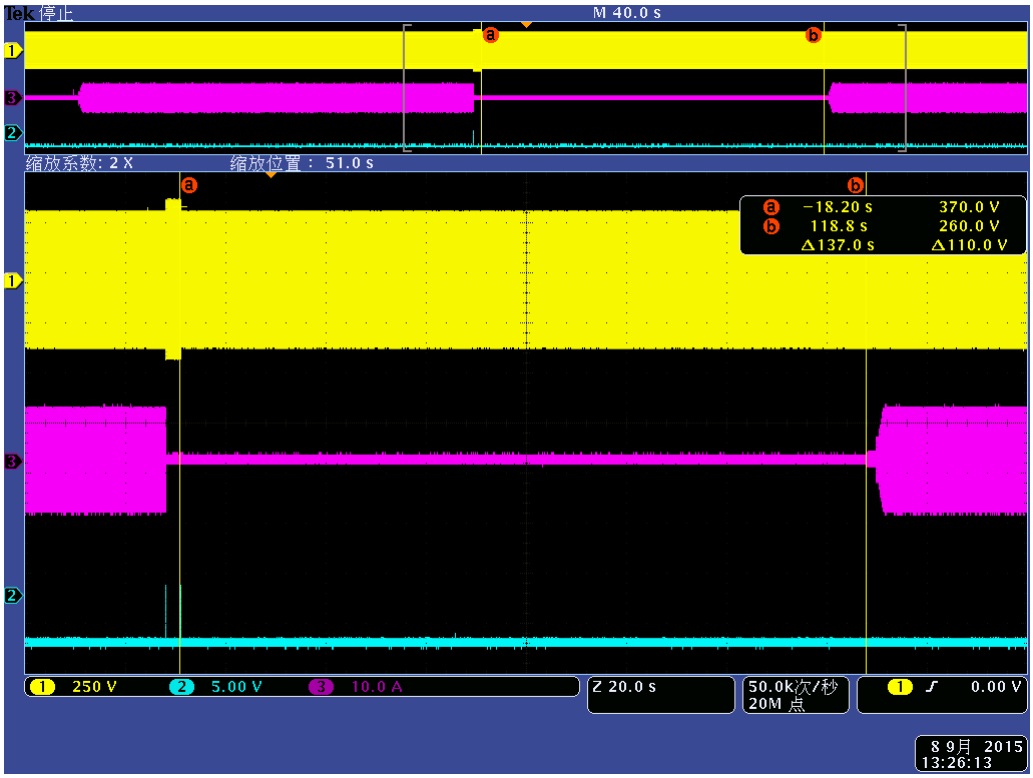
Reconnection Under Voltage second Level

Reconnection Over Voltage second Level



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5.2.2 Frequency monitoring				P
IEC 61727 **The grid-connected inverter regulations of the Metropolitan Electricity Authority (MEA)**				
	Under Frequency		**Over Frequency**	
Parameter		Frequency [Hz]		Frequency [Hz]
Output Voltage		U_N		U_N
Set value		**48,90Hz**		**51,10Hz**
Measured trip value		48,90		51,11
		48,90		51,11
		48,90		51,11
		48,90		51,11
		48,90		51,11
		Time [ms]		Time [ms]
Limit		**<= 100ms**		**<= 100ms**
Trip time	49,40 Hz to 48,40 Hz	75,0	50,60 Hz to 51,60 Hz	77,0
		89,0		79,0
		73,0		79,0
		93,0		79,0
		91,0		85,0
Reconnection time	at least 120s	137,0s	at least 120s	137,0s

Note:
The frequency which inverter stops feeding power to electrical system in each test must be in the range of the frequency trip setting +/- 0,1Hz and the time it takes to cut off the power must be within 0.1 second.

It was measured at a continuous change of frequency of 1Hz/s at lower, nominal and upper UN and arbitary output power. The trip value was determined manually by reducing the frequency in 10mHz steps. When the trip value is known (e.g. 49Hz), the ac-source is programmed to run from e.g. 49,50Hz to 48,50Hz with 1Hz/s. The disconnection time is calculated by the measured time minus the 500ms from 49,50Hz to 49,00Hz.

The frequency monitoring was tested according the test procedure 4.3.5 Operating frequency range test of the grid-connected inverter regulation Metropolitan Electricity Authority 2013. The frequency settings are set for the test as stated to 48,90Hz and 51,10Hz. The tests are perfomred according the IEEE 1547.1-2005, annex A). The reconnection time follows the test procedure of IEEE 1547.1, clause 5.10.

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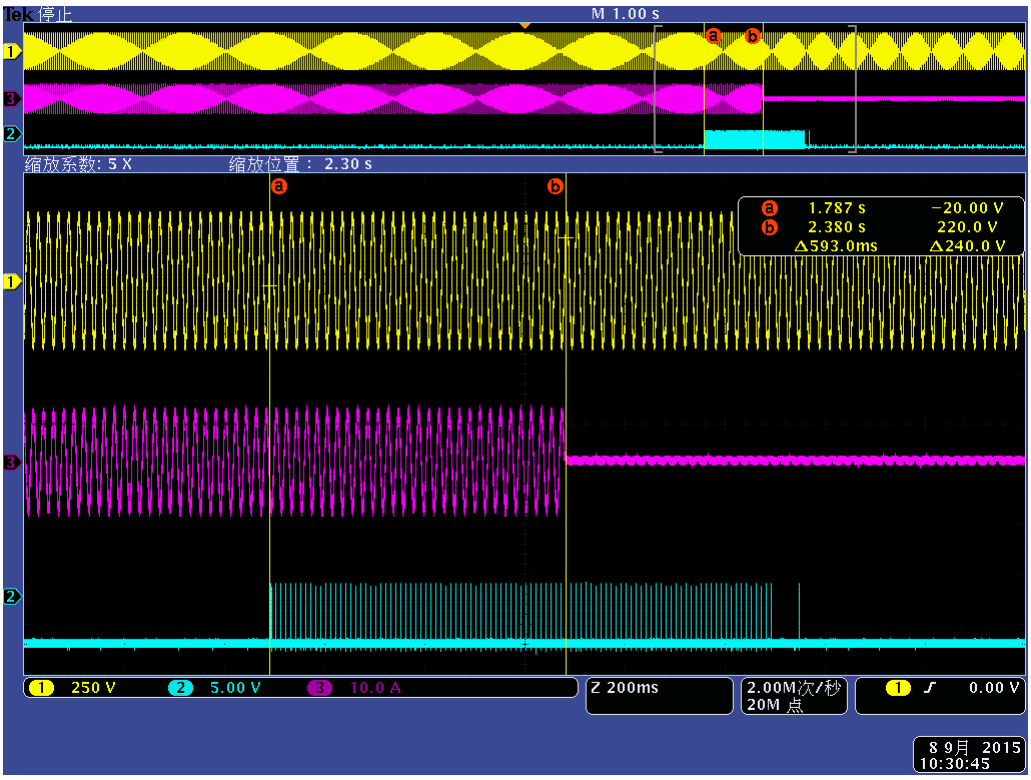
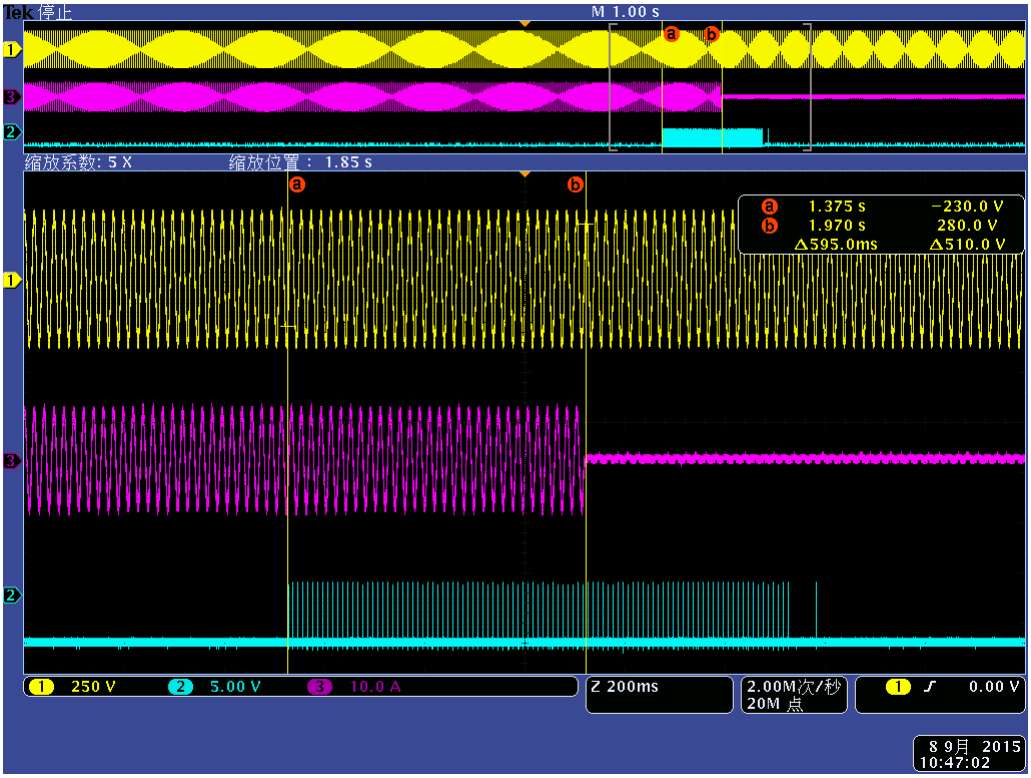
Under Frequency:



Over Frequency:



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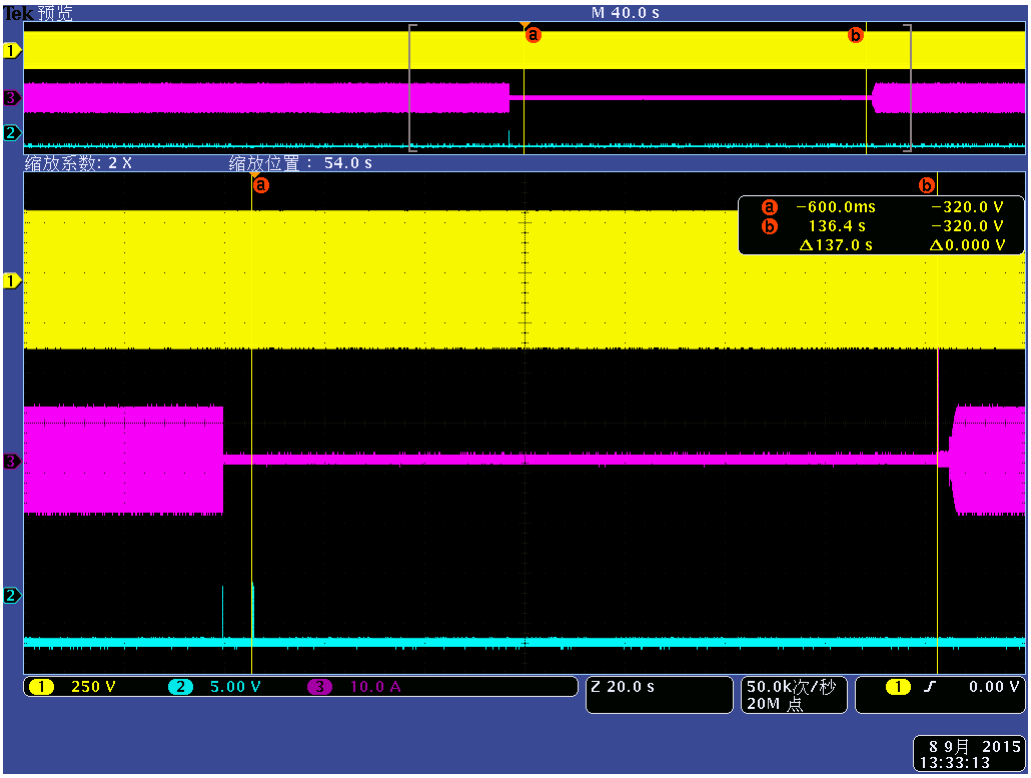
Reconnection Under Frequency:



Reconnection Over Frequency:

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6.1 Islanding protection

Test circuit and parameters

Parameter	Symbol	Units
EUT DC Input		
DC voltage	V_{DC}	V
DC Current	I_{DC}	A
DC Power	P_{DC}	W
EUT AC ouput		
AC voltage	V_{EUT}	V
AC current	I_{EUT}	A
Real power	P_{EUT}	W
Reactive power	Q_{EUT}	VAr
Test Load		
Resistive load current	I_R	A
Inductive load current	I_L	A
Capacitive load current	I_C	A
AC (utility) power source		
Utility real power	P_{AC}	W
Utility reactive power	Q_{AC}	VAr
Utility current	I_{AC}	A

Block diagram test circuit IEC 62116:2008

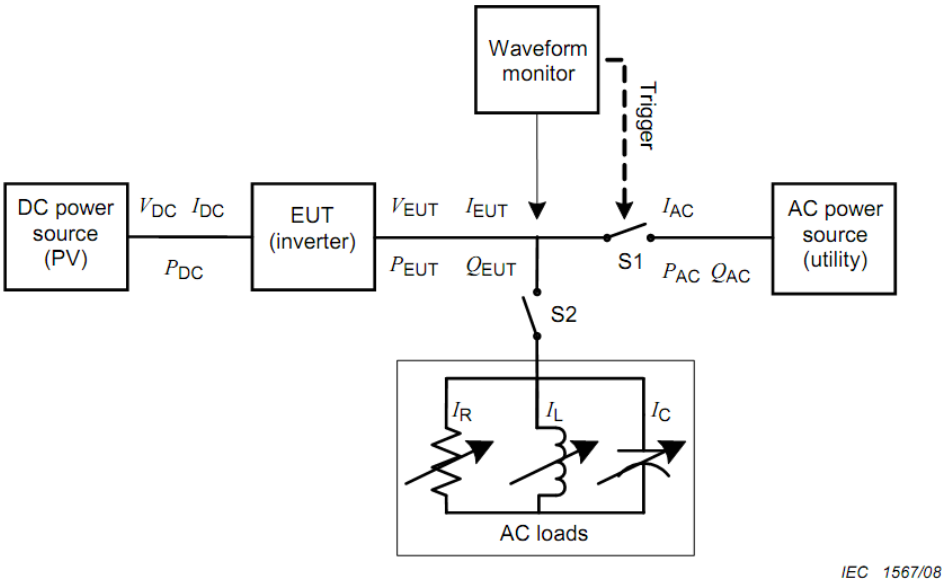

Figure 1 – Test circuit for islanding detection function in a power conditioner (inverter)

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6.1 Islanding protection according table 6 - Load imbalance (real, reactive load) for test condition A (EUT ouput = 100%)									P
Test conditions			Frequency: 50+/-0,1Hz U_N=230+/-3Vac Distortion factor of chokes < 2% Quality = 1						
Disconnection limit			0,3s deviations for MEA						
No	P_{EUT} 1) [% of EUT rating]	Reactive load [% of Q_L in 6.1.d) 1]	P_{AC} 2) [% of nominal]	Q_{AC} 3) [% of nominal]	P_{EUT} [W per phase]	V_{DC} [V]	Q_f [1]	Run on Time [ms]	Remarks 5)
1	100	100	0	0	4598	420	1,00	110,4	BL
2	100	100	-10	-10	4134	420	0,90	50,4	IB
3	100	100	-10	-5	4131	420	0,95	76,8	IB
4	100	100	-10	0	4137	420	1,00	108,8	IB
5	100	100	-10	+5	4140	420	1,05	80,0	IB
6	100	100	-10	+10	4132	420	1,09	73,6	IB
7	100	100	-5	-10	4371	420	0,90	62,6	IB
8	100	100	-5	-5	4370	420	0,95	89,6	IB
9	100	100	-5	0	4365	420	1,00	103,6	IB
10	100	100	-5	+5	4375	420	1,05	83,6	IB
11	100	100	-5	+10	4368	420	1,09	84,6	IB
12	100	100	0	-10	4596	420	0,90	57,6	IB
13	100	100	0	-5	4592	420	0,95	88,6	IB
14	100	100	0	+5	4595	420	1,05	69,6	IB
15	100	100	0	+10	4593	420	1,10	66,6	IB
16	100	100	+5	-10	4825	420	0,90	70,6	IB
17	100	100	+5	-5	4821	420	0,95	88,6	IB
18	100	100	+5	0	4819	420	1,00	71,6	IB
19	100	100	+5	+5	4815	420	1,05	78,6	IB
20	100	100	+5	+10	4830	420	1,10	82,6	IB
21	100	100	+10	-10	5050	420	0,91	62,6	IB
22	100	100	+10	-5	5020	420	0,94	80,6	IB
23	100	100	+10	0	5034	420	1,00	100,6	IB
24	100	100	+10	+5	5045	420	1,06	96,6	IB
25	100	100	+10	+10	5043	420	1,09	67,6	IB

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Parameter at 0% per phase	L= 36,62 mH	R= 11,50 Ω	C= 276,93 µF

Note:
RLC is adjusted to min. +/-1% of the inverter rated output power
1) P_{EUT}: EUT output power
2) P_{AC}: Real power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
3) Q_{AC}: Reactive power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
4) Fundamental of I_{AC} when RLC is adjusted
5) BL: Balance condition, IB: Imbalance condition.

Condition A:
EUT output power PEUT = Maximum 6)
EUT input voltage 6) = >90% of rated input voltage range

6) Maximum EUT output power condition should be achieved using the maximum allowable input power. Actual output power may exceed nominal rated output.
7) Based on EUT rated input operating range. For example, If range is between X volts and Y volts, 90 % of range =X + 0,75 × (Y – X). Y shall not exceed 0,8 × EUT maximum system voltage (i.e., maximum allowable array open circuit voltage). In any case, the EUT should not be operated outside of its allowable input voltage range.

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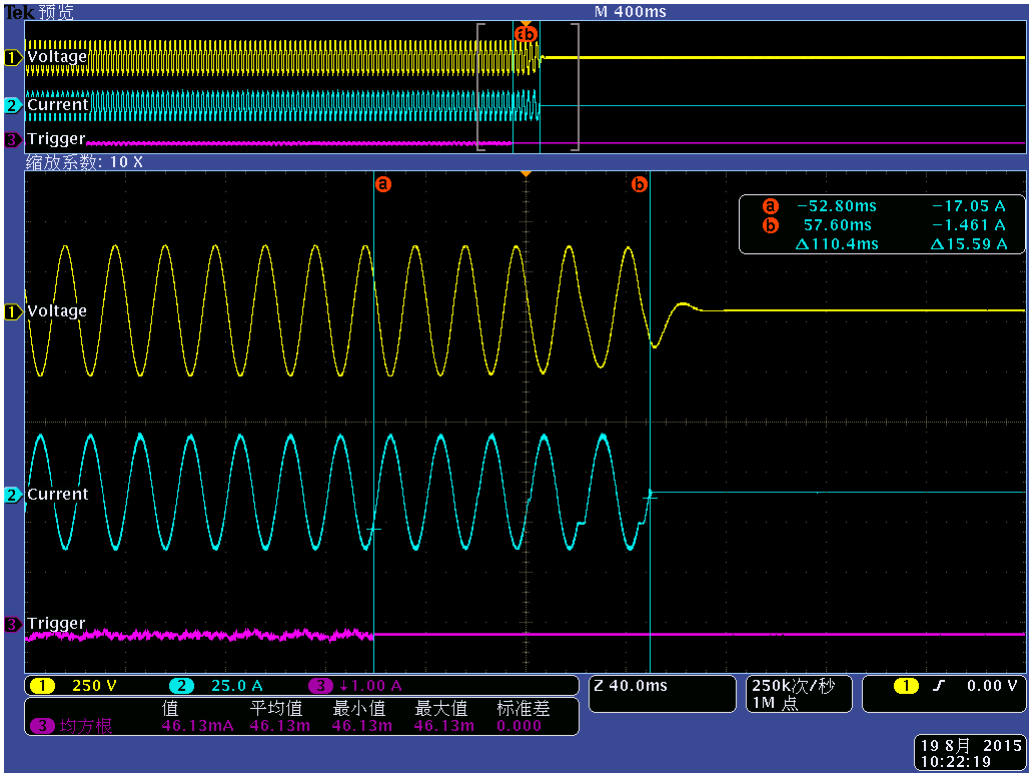
Disconnection at No. 1 of P_{AC} 0% and Q_{AC} 0% reactive load



Note:
Green waveform: the current I_{AC} pass through S1 of Figure 0. The fundamental(50Hz) component of I_{AC} at P_{AC} 0% and Q_{AC} 0% is 46,1mA.
When the switch S1 was opened, the current I_{AC} feed into the gird(AC power source) will turn to absolute zero.

CH1: EUT current
CH2: Gird voltage
CH4: I_{AC} current into the grid

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6.1 Islanding protection according Table 7 – Load imbalance (reactive load) for test condition B (EUT output = 50 % – 66 %)	**P**

Test conditions	Frequency: 50+/-0,1Hz U_N=230+/-3Vac Distortion factor of chokes < 2% Quality =1
Disconnection limit	0,3s deviations for MEA

No	P_{EUT} [1] [% of EUT rating]	Reactive load [% of Q_L in 6.1.d) 1]	P_{AC} [2] [% of nominal]	Q_{AC} [3] [% of nominal]	P_{EUT} [W per phase]	V_{DC} [V]	Q_f [1]	Run on Time [ms]	Remarks [5]
1	66	66	0	-5	3055	285	0,95	70,6	IB
2	66	66	0	-4	3065	285	0,96	81,6	IB
3	66	66	0	-3	3053	285	0,97	90,4	IB
4	66	66	0	-2	3067	285	0,98	90,4	IB
5	66	66	0	-1	3072	285	0,99	133,4	IB
6	66	66	0	0	3068	285	1,00	115,4	BL
7	66	66	0	1	3049	285	1,01	89,4	IB
8	66	66	0	2	3075	285	1,02	106,4	IB
9	66	66	0	3	3069	285	1,03	127,4	IB
10	66	66	0	4	3073	285	1,04	98,4	IB
11	66	66	0	5	3070	285	1,05	76,4	IB

Parameter at 0% per phase	L= 55,49 mH	R= 17,42 Ω	C= 182,77 µF

Note:
RLC is adjusted to min. +/-1% of the inverter rated output power
[1] P_{EUT}: EUT output power
[2] P_{AC}: Real power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
[3] Q_{AC}: Reactive power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
[4] Fundamental of I_{AC} when RLC is adjusted
[5] BL: Balance condition, IB: Imbalance condition.
Condition B:
EUT output power P_{EUT} = 50 % – 66 % of maximum
EUT input voltage [6] = 50 % of rated input voltage range, ±10 %
[6] Based on EUT rated input operating range. For example, If range is between X volts and Y volts, 50 % of range =X + 0,5 × (Y – X). Y shall not exceed 0,8 × EUT maximum system voltage (i.e., maximum allowable array open circuit voltage). In any case, the EUT should not be operated outside of its allowable input voltage range.

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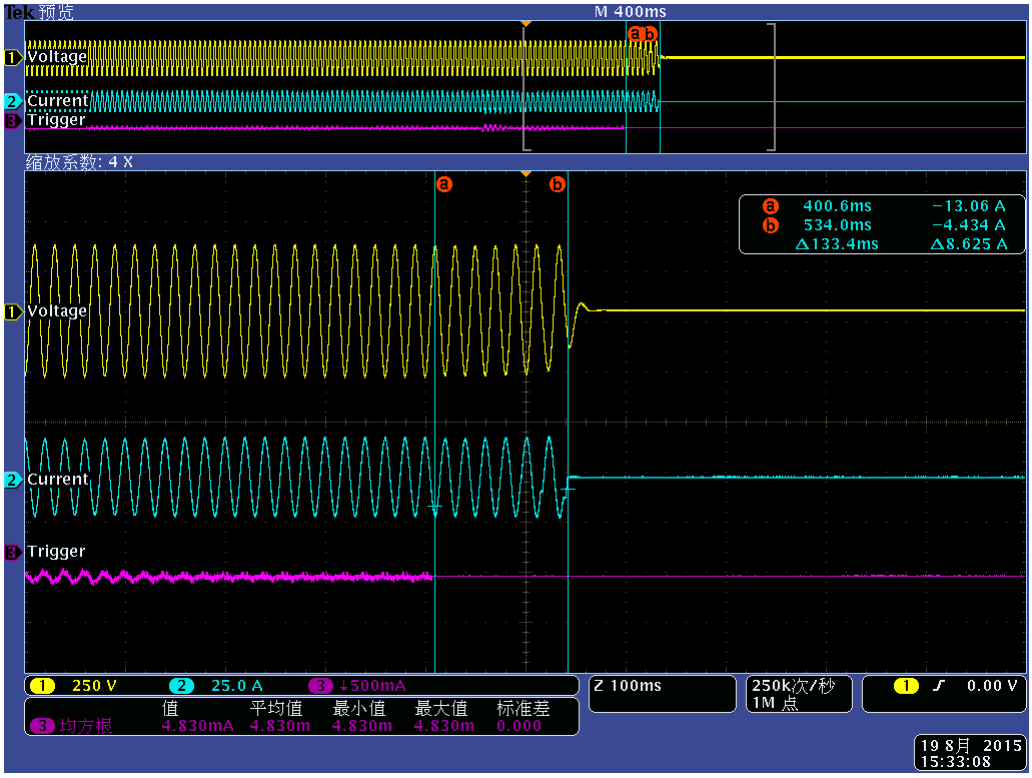
Disconnection at No. 5 of P_{AC} 0% and Q_{AC} -1% reactive load



Note:

Green waveform: the current I_{AC} pass through S1 of Figure 5. The fundamental(50Hz) component of I_{AC} at P_{AC} 0% and Q_{AC} -1% is 4,8mA.
When the switch S1 was opened, the current I_{AC} feed into the gird(AC power source) will turn to absolute zero.

CH1: EUT current
CH2: Gird voltage
CH3: I_{AC} current into the grid

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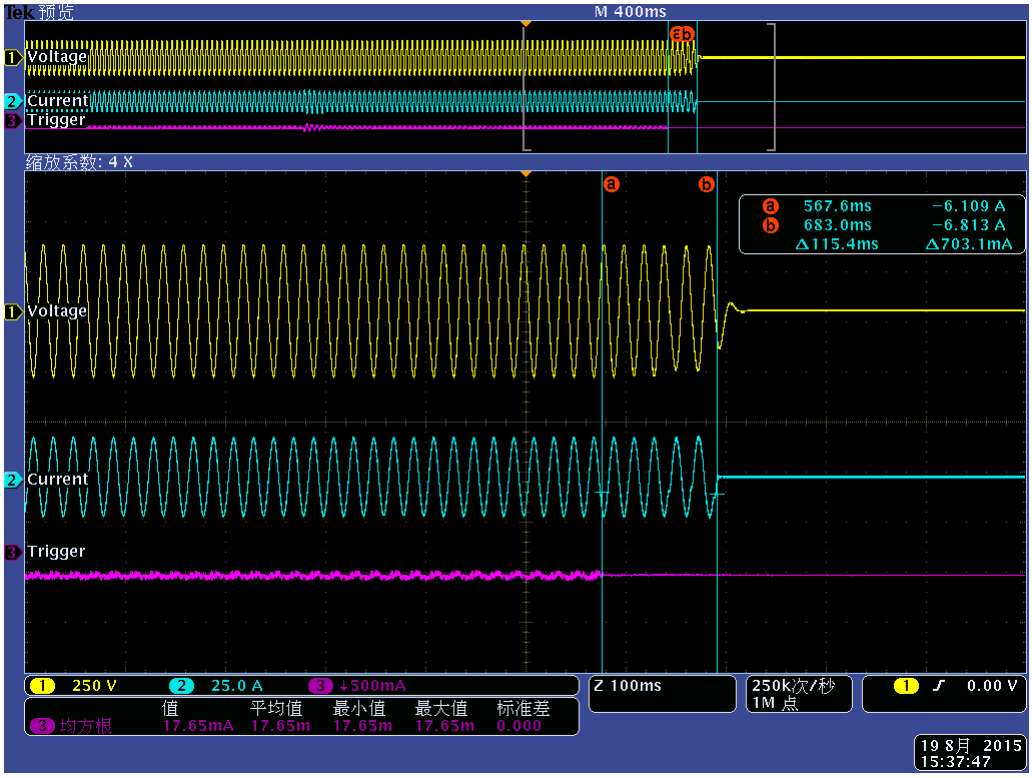
Disconnection at No. 6 of P_{AC} 0% and Q_{AC} 0% reactive load



Note:

Green waveform: the current I_{AC} pass through S1 of Figure 6. The fundamental(50Hz) component of I_{AC} at P_{AC} 0% and Q_{AC} 0% is 17,7mA.
When the switch S1 was opened, the current I_{AC} feed into the gird(AC power source) will turn to absolute zero.

CH1: EUT current
CH2: Gird voltage
CH3: I_{AC} current into the grid

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6.1 Islanding protection according Table 7 – Load imbalance (reactive load) for test condition C (EUT output = 25 % – 33 %)	**P**

Test conditions	Frequency: 50+/-0,1Hz U_N=230+/-3Vac Distortion factor of chokes < 2% Quality =1
Disconnection limit	0,3s deviations for MEA

No	P_{EUT} [1] [% of EUT rating]	Reactive load [% of Q_L in 6.1.d) 1]	P_{AC} [2] [% of nominal]	Q_{AC} [3] [% of nominal]	P_{EUT} [W per phase]	V_{DC} [V]	Q_f [1]	Run on Time [ms]	Remarks [5]
1	33	33	0	-5	1527	150	0,95	83,6	IB
2	33	33	0	-4	1532	150	0,96	79,6	IB
3	33	33	0	-3	1521	150	0,97	72,6	IB
4	33	33	0	-2	1532	150	0,98	82,6	IB
5	33	33	0	-1	1526	150	0,99	97,6	IB
6	33	33	0	0	1528	150	1,00	136,4	BL
7	33	33	0	1	1546	150	1,01	120,6	IB
8	33	33	0	2	1539	150	1,02	123,6	IB
9	33	33	0	3	1542	150	1,03	88,6	IB
10	33	33	0	4	1535	150	1,04	95,6	IB
11	33	33	0	5	1532	150	1,05	80,6	IB

Parameter at 0% per phase	L= 110,98 mH	R= 34,85 Ω	C= 91,39 µF

Note:
RLC is adjusted to min. +/-1% of the inverter rated output power
[1] P_{EUT}: EUT output power
[2] P_{AC}: Real power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
[3] Q_{AC}: Reactive power flow at S1 in Figure 1. Positive means power from EUT to utility. Nominal is the 0 % test condition value.
[4] Fundamental of I_{AC} when RLC is adjusted
[5] BL: Balance condition, IB: Imbalance condition.
Condition B:
EUT output power PEUT = 25 % – 33 % [6] of maximum
EUT input voltage [7] = <10 % of rated input voltage range
[6] Or minimum allowable EUT output level if greater than 33 %.
[7] Based on EUT rated input operating range. For example, If range is between X volts and Y volts, 10 % of range =X + 0,2 × (Y – X). Y shall not exceed 0,8 × EUT maximum system voltage (i.e., maximum allowable array open circuit voltage). In any case, the EUT should not be operated outside of its allowable input voltage range.

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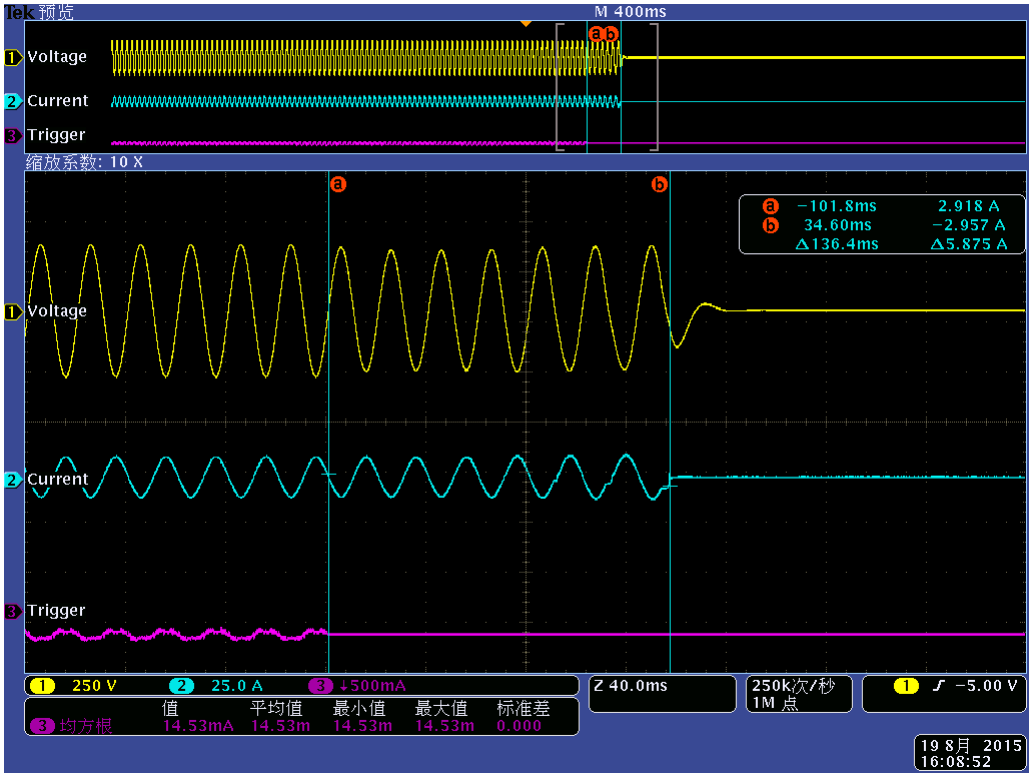
Disconnection at No. 6 of P_{AC} 0% and Q_{AC} 0% reactive load



Note:
Green waveform: the current I_{AC} pass through S1 of Figure 6. The fundamental(50Hz) component of I_{AC} at P_{AC} 0% and Q_{AC} 0% is 14,5mA.
When the switch S1 was opened, the current I_{AC} feed into the gird(AC power source) will turn to absolute zero.

CH1: EUT current
CH2: Gird voltage
CH3: I_{AC} current into the grid

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Annex 1

EMC Test Report

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Technical Compliance Statement

CE

Ref. No.: ACWE-RC130109 (ACWE-G1306031)

For the following equipment

Applicant	:	**Omnik New Energy Co., Ltd**
Manufacturer	:	**Omnik New Energy Co., Ltd**
Product	:	**Solar Inverter**
Model Number	:	**(1) Omniksol-5k-TL2 (2) Omniksol-4k-TL2 (3) Omniksol-3k-TL2**
Brand	:	**Omnik**

We, **AUDIX Technology (Wujiang) Co., Ltd. EMC Dept.** hereby certify that the above products has been tested by us with the listed standards and found in compliance with the council EMC directive 2004/108/EC. It is possible to use CE marking to demonstrate the compliance with this EMC Directive. The test data & results are issued on the EMC test report No. **ACWE-E1307007.**

Emission: **EN 61000-6-3:2007**
EN 61000-3-11:2000 and EN 61000-3-12:2011

Immunity: **EN 61000-6-2:2005**
(IEC 61000-4-2:2008, IEC 61000-4-3:2008,
IEC 61000-4-4:2004+Corr.1:2006+Corr.2:2007, IEC 61000-4-5:2005,
IEC 61000-4-6:2008, IEC 61000-4-8:2009, IEC 61000-4-11:2004)

Jul.12, 2013

Allen Wang

Allen Wang / Deputy General Manager
AUDIX Technology (Wujiang) Co., Ltd. EMC Dept.

The statement is based on a single evaluation of one sample of the above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab logo.

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EMC TEST REPORT
For

Omnik New Energy Co.,Ltd

Solar Inverter

Model No. : (1) Omniksol-5k-TL2 (2) Omniksol-4k-TL2 (3) Omniksol-3k-TL2
Brand : Omnik

Prepared for

Omnik New Energy Co.,Ltd
Xinghu Road No.218 BioBAY Park A4-414, 215213 Suzhou China

Prepared by

Audix Technology (Wujiang) Co., Ltd. EMC Dept.
No.1289 Jiangxing East Road, the Eastern Part of Wujiang Economic Development Zone
Jiangsu China 215200

Tel.: +86-512-63403993
Fax: +86-512-63403339

Report Number	:	ACWE-E1307007
Date of Test	:	Jun.27~Jul.03, 2013
Date of Report	:	Jul.08, 2013

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APPENDIX Photos of EUT

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TEST REPORT VERIFICATION

Applicant	:	Omnik New Energy Co.,Ltd
Manufacturer	:	Omnik New Energy Co.,Ltd
EUT Description	:	Solar Inverter
(A) Model No.	:	(1) Omniksol-5k-TL2 (2) Omniksol-4k-TL2 (3) Omniksol-3k-TL2
(B) Brand	:	Omnik
(C) Test Voltage	:	AC 230V, 50Hz

Applicable standards:

Emission: **EN 61000-6-3:2007**
EN 61000-3-11:2000 and EN 61000-3-12:2011

Immunity: **EN 61000-6-2:2005**
(IEC 61000-4-2:2008, IEC 61000-4-3:2008,
IEC 61000-4-4:2004+Corr.1:2006+Corr.2:2007, IEC 61000-4-5:2005,
IEC 61000-4-6:2008, IEC 61000-4-8:2009, IEC 61000-4-11:2004)

The device described above is tested by Audix Technology (Wujiang) Co., Ltd. EMC Dept. to determine the Maximum emission levels emanating from the device, its ensured severity levels, and performance criterion. This test report contains the measurement results, and Audix Technology (Wujiang) Co., Ltd. EMC Dept. assumes full responsibility for the accuracy and completeness of these measurements. Also, this report shows that the EUT is technically compliance with the requirements of EN 61000-6-3、-2and EN 6100-3-11、-12 standards.

This report applies to above tested sample only and shall not be reproduced in part without written approval of Audix Technology (Wujiang) Co., Ltd. EMC Dept.

Date of Test: Jun.27~Jul.03, 2013 Date of Report: Jul.08, 2013

Prepared by : Emma Hu
(Emma Hu/Assistant Administrator)

Reviewer : Jingo Lin
(Jingo Lin/ Section Manager)

Approved & Authorized Signer : Allen Wang
(Allen Wang/Deputy General Manager)

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1 SUMMARY OF STANDARDS AND RESULTS

1.1 Description of Standards and Results

The EUT has been tested according to the applicable standards and test results are referred as below.

EMISSION (EN 61000-6-3:2007)			
Description of Test Item	**Standard**	**Limits**	**Results**
Conducted disturbance at main terminal	EN 61000-6-3:2007	Class B	PASS
Conducted common mode disturbance at telecommunication port	EN 61000-6-3:2007	Class B	Note
Radiated disturbance	EN 61000-6-3:2007	Class B	PASS
Harmonic current emissions	EN 61000-3-11:2000	Stage 2	PASS
Voltage fluctuations & flicker	EN 61000-3-12:2011	P_{st}=1 dc(%)=3.3% dMax.(%)=4% d(t)>3.3%=500ms	PASS
IMMUNITY (EN 61000-6-2:2005)			
Description of Test Item	**Basic Standard**	**Performance Criteria**	**Results**
Electrostatic discharge (ESD)	IEC 61000-4-2:2008	B	PASS
Radio-frequency, Continuous radiated disturbance	IEC 61000-4-3:2008	A	PASS
Electrical fast transient (EFT)	IEC 61000-4-4:2004 +Corr.1:2006+Corr.2:2007	B	PASS
Surge	IEC 61000-4-5:2005	B	PASS
Radio-frequency, Continuous conducted disturbance	IEC 61000-4-6:2008	A	PASS
Power frequency magnetic field	IEC 61000-4-8:2009	A	PASS
Voltage dips, >95% reduction	IEC 61000-4-11:2004	B	PASS
Voltage dips, 30% reduction		C	PASS
Voltage interruptions		C	PASS
Note : Due to the EUT doesn't have telecommunication port; this test item of the telecommunication terminal is not applicable for it.			

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1.2 Description of Performance Criteria

The variety and the diversity of the apparatus within the scope of this standard makes it difficult to define precise criteria for the evaluation of the immunity test results.

If, as a result of the application of the tests defined in this standard, the apparatus becomes dangerous or unsafe, the apparatus shall be deemed to have failed the test.

A functional description and a definition of performance criteria, during or as a consequence of the EMC testing, shall be provided by the manufacturer and noted in the test report, based on one of the following criteria for each test as specified in Tables 1 to 4.

1.2.1 Performance criterion A

The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.

1.2.2 Performance criterion B

The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.

1.2.3 Performance criterion C

Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls.

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2 GENERAL INFORMATION

2.1 Description of Device (EUT)

Product	:	Solar Inverter
Model Number	:	(1) Omniksol-5k-TL2 (2) Omniksol-4k-TL2 (3) Omniksol-3k-TL2
Brand	:	Omnik
Applicant	:	Omnik New Energy Co.,Ltd Xinghu Road No.218 BioBAY Park A4-414, 215213 Suzhou China
Manufacturer	:	Omnik New Energy Co.,Ltd Xinghu Road No.218 BioBAY Park A4-414, 215213 Suzhou China
Date of Receipt of Sample	:	Jun.27, 2013
Date of Test	:	Jun.27~Jul.03, 2013
Oscillator Frequency		10MHz,14.7456MHz
Working Frequency	:	20KHz

Remark:
1. The differences among these models are showed as below.

M/N / Differences	Omniksol-3k-TL2	Omniksol-4k-TL2	Omniksol-5k-TL2
Electrolytic capacitor	CA3、CA4、CA5、CA11、CA12、CA13	CA3、CA4、CA5、CA6、CA11、CA12、CA13、CA14	CA3、CA4、CA5、CA6、CA7、CA11、CA12、CA13、CA14、CA15

2. Besides these, the EUT's electronic circuit design and PCB layout are all the same, so the highest output power of EUT "Omniksol-5k-TL2" was selected to estimating in this report.

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2.2 EUT's Specification under application

Type	Omniksol-3k-TL2	Omniksol-4k-TL2	Omniksol-5k-TL2
Input (DC)			
Max. PV-Generator Power (W)	3400	4600	5200
Max. DC Voltage (V)	590	590	590
Rated Voltage (V)	360	360	360
MPPT DC Voltage Range (V)	120-500	120-500	120-500
Turn off DC Voltage (V)	90	90	90
Max. DC Current (A)	12*2	16*2	18*2
Nominal DC Current (A)	10*2	13*2	14*2
Number of DC Connection	2	2	2
Max. Input Power, each MPPT (W)	2000	2600	3000
DC-Connection	MC4	MC4	MC4
Number of MPP trackers	2	2	2
Turn on Power (W)	10	10	10
Output (AC)			
Max. AC Power (VA)	3300	4400	5000
Nominal AC Power (W)	3000	4000	4600
Max. AC Current (A)	14.4	19.0	22.0
Nominal AC Current (A)	13.0	17.4	20.0
Grid Voltage Range	185V…276V(depending on regional settings)	185V…276V(depending on regional settings)	185V…276V(depending on regional settings)
Rated Frequency	50Hz and 60Hz	50Hz and 60Hz	50Hz and 60Hz
Power Factor	0.95 capacitive…0.95 inductive	0.95 capacitive…0.95 inductive	0.95 capacitive…0.95 inductive
THD(AC Output current)	<2%	<2%	<2%
AC Connector	Plug-in connector	Plug-in connector	Plug-in connector

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2.3 Operating Condition of EUT

2.3.1 Set up the EUT as showed each respective block diagram of test setup.

2.3.2 Turn on the power of the EUT.

2.3.2.1 Adjust the output of the DC power supply which made the EUT working in Full load and Half load for EMI test, 10% load for EMS test.

2.3.3 Details as below:

Input / Load	Voltage (V)	Current (A)
Full Load	370V	12.4A
Half Load	370V	6.2A
10% Load	370V	1.25A

2.4 Tested Supporting System Details

2.4.1 DC Power Supply

Manufacturer	:	Topcon
Model Number	:	TC.P.10.800.400.PV.HMI
Input Power	:	AC: 3*400V, 3*21A
Output	:	0~10kW, 0~800Vdc, 0~16A

2.5 Description of Test Facility

Name of Firm #1	:	**Audix Technology (Wujiang) Co., Ltd. EMC Dept.**
Site Location #1	:	No. 1289 Jiangxing East Road, the Eastern Part of Wujiang Economic Development Zone Jiangsu China 215200
Test Facilities	:	**No.1 10m semi-anechoic chamber** **No. 2 conducted shielding enclosure** **The Complex Immunity Test Room** **RS&CS Test Room**
NVLAP Lab Code	:	200786-0 (NVLAP is a NATA accredited body under Mutual Recognition Agreement) Valid date until on Sep.30, 2013

2.6 Measurement Uncertainty

Test Item	Range Frequency	Uncertainty
Conducted Disturbance Measurement at mains	0.15MHz ~ 30MHz	± 2.36dB
At 10m Semi-Anechoic Chamber		
Radiated Disturbance Measurement	30MHz ~ 1000MHz	± 3.06dB (Horizontal)
		± 3.10dB (Vertical)

Remark：Uncertainty = $ku_c(y)$

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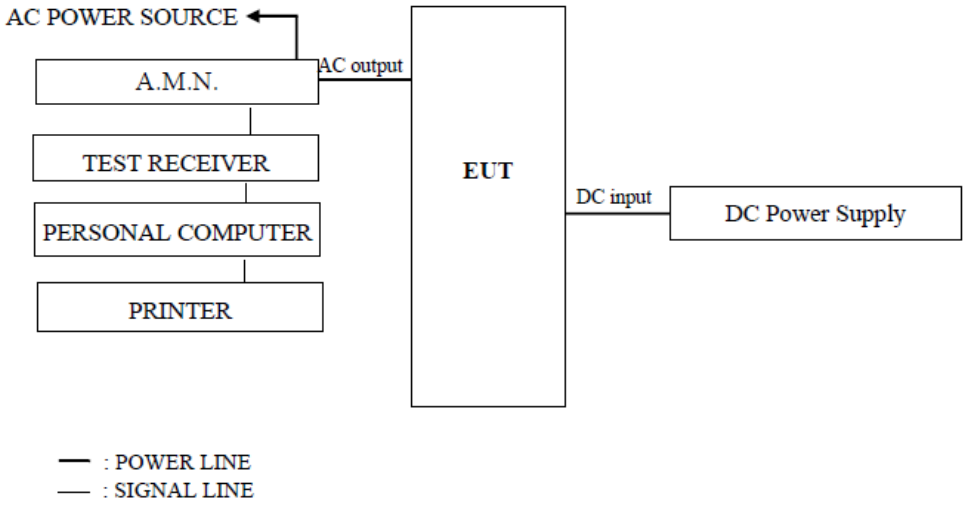
3 CONDUCTED DISTURBANCE MEASUREMENT

3.1 Test Equipment

The following test equipment was used during the conducted emission measurement：
For AC Mains Port

Item	Type	Manufacturer	Model No.	Serial No.	Last Cal.	Next Cal.
1.	Test Receiver	R & S	ESCI	100352	2013-01-05	2014-01-04
2.	A.M.N.	Schwarzbeck	NNLK 8129	8129-164	2012-08-06	2013-08-05
3.	L.I.S.N.	Kyoritsu	KNW-407	8-1793-4	2012-08-06	2013-08-05
4.	50ohm Terminator	Tektronis	MS4630B	001-con	2013-01-05	2014-01-04
5.	RF Cable	Harbour Industries	RG400	002	2013-03-24	2014-03-23

3.2 Block Diagram of Test Setup



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3.3 Limits for Conducted Disturbance Voltage

Frequency	Maximum RF Line Voltage	
	Quasi-Peak Level	Average Level
EN61000-6-3		
0.15MHz ~ 0.5MHz	66 ~ 56 dBμV	56 ~ 46 dBμV
0.5MHz ~ 5MHz	56 dBμV	46 dBμV
5MHz ~ 30MHz	60 dBμV	50 dBμV

Remark 1. If the average limit is met when using a Quasi-Peak detector, the EUT shall be deemed to meet both limits and measurement with the average detector is unnecessary.
2. The lower limit applies at the band edges.

3.4 Test Procedure

The measuring process is according to EN 61000-6-3:2007, Class B and laboratory internal procedure TKC-301-015.

In the conducted emission measurement, the EUT and all peripheral devices were set up on a non-metallic table which was 0.8 meters height above the ground plane, and 0.4 meters far away from the vertical plane. The EUT was powered by AC mains through Artificial Mains Network (A.M.N), other peripheral devices were powered by AC mains through the second Line Impedance Stabilization Network (L.I.S.N). For the measurement, the A.M.N measuring port was terminated by 50Ω measuring equipment and the second L.I.S.N measuring port was terminated by a 50Ω resistive load. All measurements were done on the phase and neutral line of the EUT's power cord. All cables or wires placement were verified to find out the maximum emission.

The bandwidth of measuring receiver was set at 9 kHz.

The required frequency band (0.15 MHz ~ 30 MHz) was pre-scanned with peak detector; the final measurement was measured with quasi-peak detector and average detector.

The emission level is calculated automatically by the test system which uses the following equation:

Emission level (dBμV) = Meter-Reading (dBμV) + A.M.N factor (dB) + Cable loss (dB).
(Cable loss include pulse limiter loss)

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3.5 Measurement Results

PASSED.

(All the emissions not reported below are too low against the prescribed limits.)
EUT with the following test modes were performed during this section testing and all the test results are listed in next pages.

Test Date：Jul.03, 2013 Temperature：21.2℃ Humidity：51%

Item	Test Condition	Reference Test Data No.	
		Neutral	Line
1	Half Load	# 2	# 1
2	**Full Load**	**# 4**	**※# 3**

NOTE 1 - '※'means the worst test mode.

NOTE 2 - The worst emission is detected at 12.72 MHz with emission level of 49.92 dB (μV) (limit is 50.00dB (μV)) with AV detector, when the Line of the EUT is connected to A.M.N.

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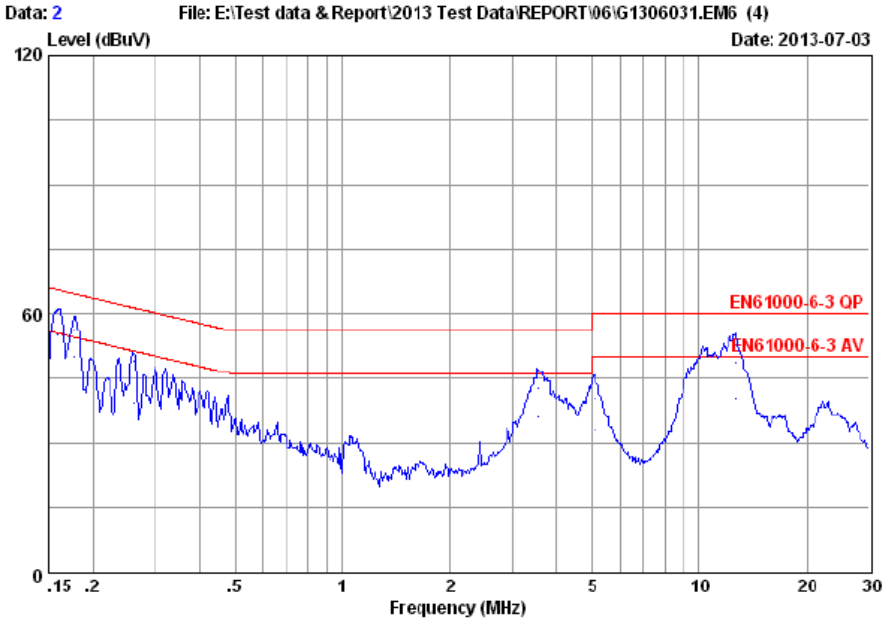

Site no.	: No.2 Conducted shielding Enclosure	Data no.	: 2
AMN/LISN	: NNLK8129-N-1208	Phase	: NEUTRAL
Limit	: EN61000-6-3 QP		
Env. / Ins.	: 21.2C&51%/ESCI	Engineer	: KM Tong
EUT	: Solar Inverter		
M/N	: Omniksol-5k-TL2		
Power Rating	: 230Vac/50Hz		
Test mode	: Half load		
Memo	:		

	Freq. (MHz)	AMN Factor (dB)	Cable Loss (dB)	Reading (dBuV)	Emission Level (dBuV)	Limits (dBuV)	Margin (dB)	Remark
1	0.16	0.18	10.22	50.69	61.09	65.52	4.43	QP
2	0.16	0.18	10.22	44.59	54.99	55.52	0.53	Average
3	0.18	0.18	10.26	45.71	56.15	64.58	8.43	QP
4	0.18	0.18	10.26	39.61	50.05	54.58	4.53	Average
5	0.26	0.18	10.48	28.90	39.56	51.43	11.87	Average
6	0.26	0.18	10.48	38.50	49.16	61.43	12.27	QP
7	3.52	0.17	10.31	32.20	42.68	56.00	13.32	QP
8	3.52	0.17	10.31	25.60	36.08	46.00	9.92	Average
9	5.12	0.19	10.27	30.00	40.46	60.00	19.54	QP
10	5.12	0.19	10.27	22.70	33.16	50.00	16.84	Average
11	12.76	0.34	10.38	29.81	40.53	50.00	9.47	Average
12	12.76	0.34	10.38	38.21	48.93	60.00	11.07	QP

1. Emission Level= AMN Factor + Cable Loss + Reading.
2. If the average limit is met when useing a quasi-peak detector, the EUT shall be deemed to meet both limits and measurement with average detector is unnecessary.

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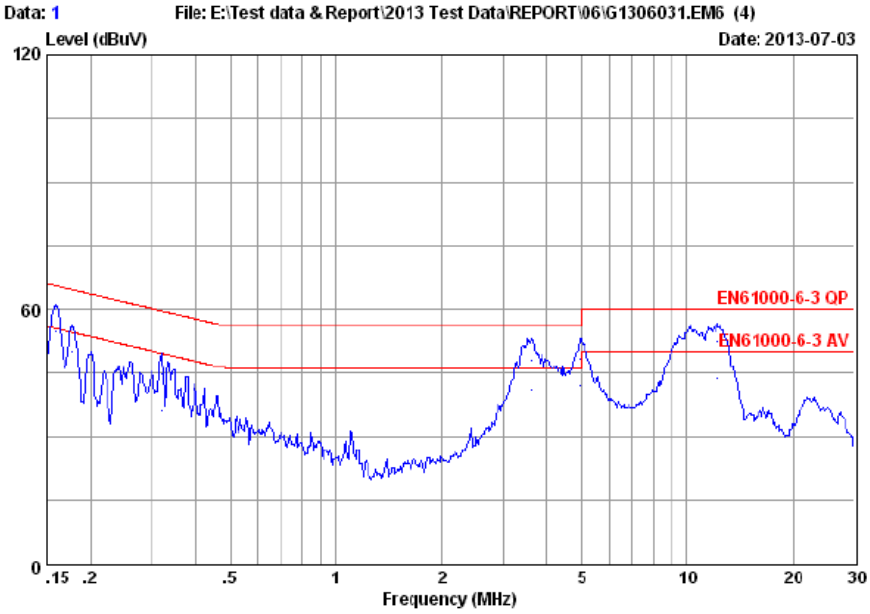

Site no.	: No.2 Conducted shielding Enclosure	Data no.	: 1
AMN/LISN	: NNLK8129-L1-1208	Phase	: LINE
Limit	: EN61000-6-3 QP		
Env. / Ins.	: 21.2C&51%/ESCI	Engineer	: KM Tong
EUT	: Solar Inverter		
M/N	: Omniksol-5k-TL2		
Power Rating	: 230Vac/50Hz		
Test mode	: Half load		
Memo	:		

	Freq. (MHz)	AMN Factor (dB)	Cable Loss (dB)	Reading (dBuV)	Emission Level (dBuV)	Limits (dBuV)	Margin (dB)	Remark
1	0.16	0.25	10.22	50.29	60.76	65.52	4.76	QP
2	0.16	0.25	10.22	44.29	54.76	55.52	0.76	Average
3	0.18	0.25	10.26	39.60	50.11	54.58	4.47	Average
4	0.18	0.25	10.26	45.90	56.41	64.58	8.17	QP
5	0.32	0.21	10.59	34.11	44.91	59.73	14.82	QP
6	0.32	0.21	10.59	25.31	36.11	49.73	13.62	Average
7	3.59	0.17	10.31	37.50	47.98	56.00	8.02	QP
8	3.59	0.17	10.31	30.50	40.98	46.00	5.02	Average
9	4.94	0.18	10.27	31.60	42.05	46.00	3.95	Average
10	4.94	0.18	10.27	37.90	48.35	56.00	7.65	QP
11	12.24	0.33	10.37	33.20	43.90	50.00	6.10	Average
12	12.24	0.33	10.37	41.80	52.50	60.00	7.50	QP

1. Emission Level= AMN Factor + Cable Loss + Reading.
2. If the average limit is met when useing a quasi-peak detector, the EUT shall be deemed to meet both limits and measurement with average detector is unnecessary.

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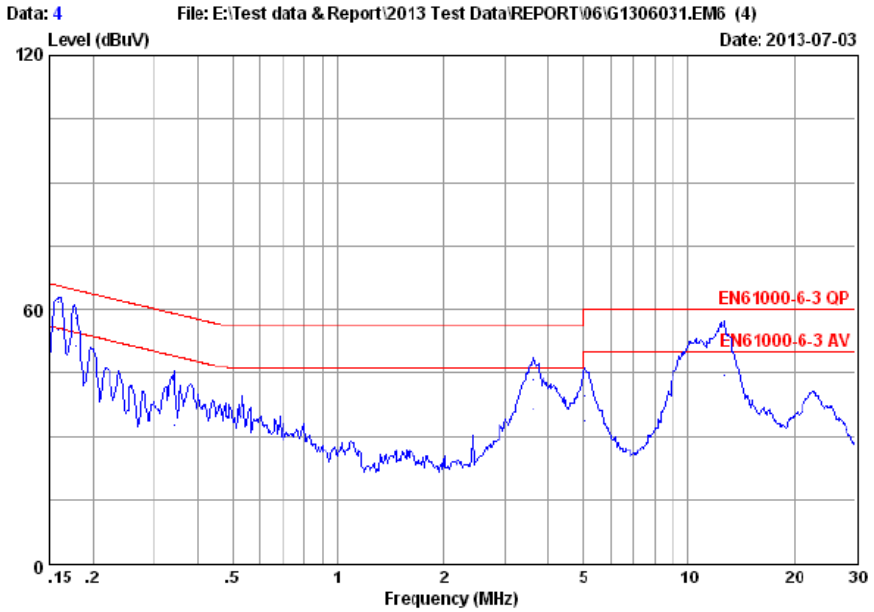

Site no. : No.2 Conducted shielding Enclosure
AMN/LISN : NNLK8129-N-1208
Limit : EN61000-6-3 QP
Env. / Ins. : 21.2C&51%/ESCI
EUT : Solar Inverter
M/N : Omniksol-5k-TL2
Power Rating : 230Vac/50Hz
Test mode : Full Load
Memo :

Data no. : 4
Phase : NEUTRAL
Engineer : KM Tong

	Freq. (MHz)	AMN Factor (dB)	Cable Loss (dB)	Reading (dBuV)	Emission Level (dBuV)	Limits (dBuV)	Margin (dB)	Remark
1	0.16	0.18	10.22	51.59	61.99	65.52	3.53	QP
2	0.16	0.18	10.22	44.89	55.29	55.52	0.23	Average
3	0.18	0.18	10.27	47.60	58.05	64.53	6.48	QP
4	0.18	0.18	10.27	41.10	51.55	54.53	2.98	Average
5	0.34	0.17	10.62	31.40	42.19	59.20	17.01	QP
6	0.34	0.17	10.62	22.00	32.79	49.20	16.41	Average
7	3.61	0.17	10.31	33.00	43.48	56.00	12.52	QP
8	3.61	0.17	10.31	26.20	36.68	46.00	9.32	Average
9	5.04	0.18	10.27	29.40	39.85	60.00	20.15	QP
10	5.04	0.18	10.27	23.30	33.75	50.00	16.25	Average
11	12.72	0.34	10.38	42.10	52.82	60.00	7.18	QP
12	12.72	0.34	10.38	33.80	44.52	50.00	5.48	Average

1.Emission Level= AMN Factor + Cable Loss + Reading.
2.If the average limit is met when useing a quasi-peak detector,the EUT shall be deemed to meet both limits and measurement with average detector is unnecessary.

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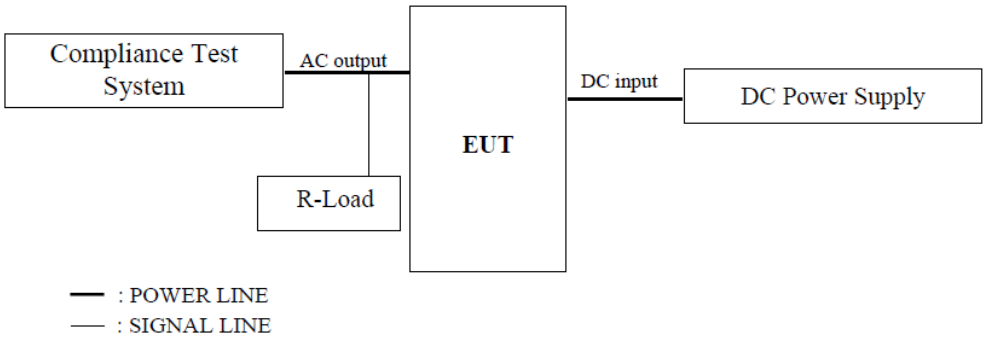
5 POWER HARMONICS AND FLICKER MEASUREMENT

5.1 Test Equipment

Item	Type	Manufacturer	Model No.	Last Cal.	Next Cal.
1.	Harmonic Flicker	TESEQ AG	CCN1000-3	2013-02-28	2014-02-27
2.	Harmonic Flicker	TESEQ AG	INA2196	2013-02-28	2014-02-27
3.	Harmonic Flicker	TESEQ AG	INA2197	2013-02-28	2014-02-27

5.2 Block Diagram of Test Setup



5.3 Test Standard

EN 61000-3-11:2000 and EN 61000-3-12:2011

5.4 Test Procedure

The measuring process is according to EN 61000-3-11:2000and EN 61000-3-12:2011 and laboratory internal procedure TKC-301-026.

5.5 Test Results

PASSED. (Complied with Stage 2 limit).
EUT with the following test modes was measured during this section testing and all the test results are listed in next page.

Test Item	Test Condition
Harmonic	33% Load
Harmonic	66% Load
Harmonic	Full Load
Flicker	Full Load

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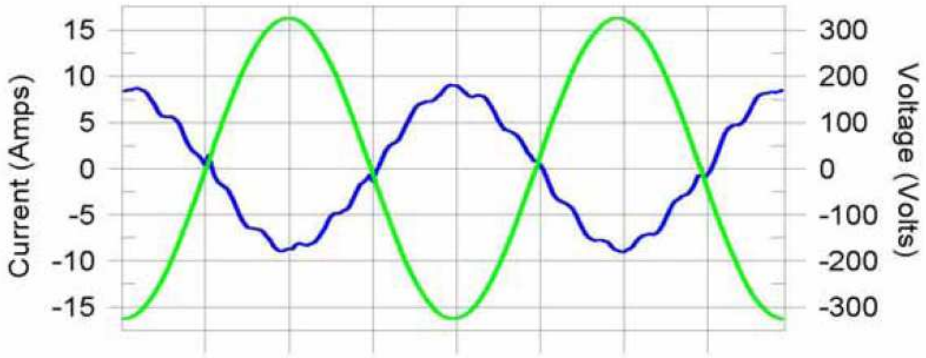
Schaffner Profline
San Diego, California

Harmonics – Per EN/IEC61000-3-12(Run time)

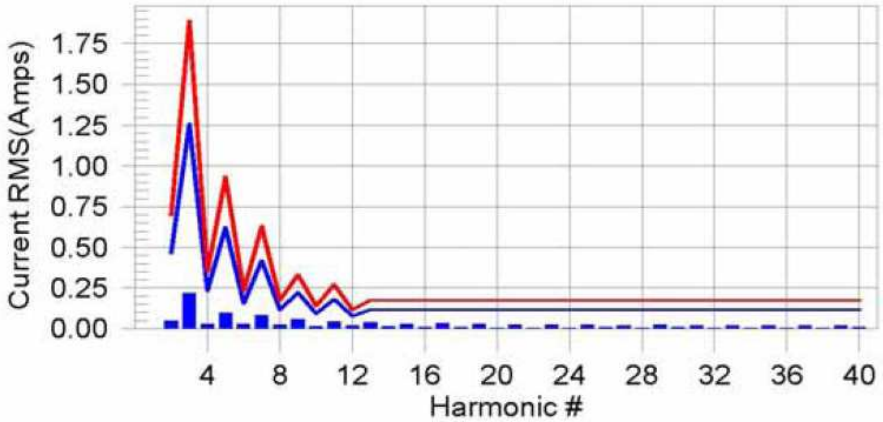
EUT: Solar Inverter **Tested by: Puck**
Test category: Stage:2 1ph & NB 3ph(Rsce=33) **Test Margin: 100**
Test date: 2013-7-2
Test duration (min): 5
Comment: M/N:Omniksol-5k-TL2 Test Mode:33% Load
Customer: Omnik New Energy Co.ltd

Test Result: Pass **Source qualification: Normal**

Current & voltage waveforms



Harmonics and Class 2 limit line **European Limits**



Test result: Pass **Worst harmonic was #13 with 31.28 % of the limit.**

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Schaffner Profline
San Diego, California

Current Test Result Summary (Run time)

EUT: Solar Inverter **Tested by: Puck**
Test category: Stage:2 1ph & NB 3ph(Rsce=33) **Test Margin: 100**
Test date: 2013-7-2
Test duration (min): 5
Comment: M/N:Omniksol-5k-TL2 Test Mode:33% Load
Customer: Omnik New Energy Co.ltd

Test Result: Pass **Source qualification: Normal**
I-THD(%): 4.758 **Limit(%): 23.000** **PWHD(%): 6.826** **PWHD Limit(%): 23.000**

Highest parameter values during test:

V_RMS (Volts):	230.11	Frequency(Hz):	50.00
I_Peak (Amps):	9.302	I_RMS (Amps):	6.015
I_1-Ref (Amps):	5.843	Crest Factor:	1.552
Power (Watts):	-1374	Power Factor:	-0.994

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.047	0.466	10.0	0.049	0.699	7.03	Pass
3	0.210	1.258	16.7	0.218	1.888	11.56	Pass
4	0.027	0.233	11.4	0.028	0.350	8.15	Pass
5	0.097	0.623	15.5	0.099	0.935	10.57	Pass
6	0.027	0.155	17.2	0.029	0.233	12.43	Pass
7	0.079	0.419	18.8	0.082	0.629	12.99	Pass
8	0.021	0.117	17.8	0.023	0.175	12.98	Pass
9	0.056	0.221	25.1	0.058	0.332	17.48	Pass
10	0.013	0.093	14.2	0.016	0.140	11.28	Pass
11	0.041	0.181	22.5	0.043	0.271	15.94	Pass
12	0.018	0.078	23.7	0.021	0.117	18.06	Pass
13	0.036	0.117	31.0	0.040	0.175	22.74	Pass
14	0.010	N/A	N/A	0.013	N/A	N/A	N/A
15	0.029	N/A	N/A	0.031	N/A	N/A	N/A
16	0.006	N/A	N/A	0.008	N/A	N/A	N/A
17	0.032	N/A	N/A	0.034	N/A	N/A	N/A
18	0.010	N/A	N/A	0.011	N/A	N/A	N/A
19	0.026	N/A	N/A	0.029	N/A	N/A	N/A
20	0.003	N/A	N/A	0.005	N/A	N/A	N/A
21	0.022	N/A	N/A	0.024	N/A	N/A	N/A
22	0.002	N/A	N/A	0.004	N/A	N/A	N/A
23	0.019	N/A	N/A	0.022	N/A	N/A	N/A
24	0.003	N/A	N/A	0.005	N/A	N/A	N/A
25	0.020	N/A	N/A	0.024	N/A	N/A	N/A
26	0.003	N/A	N/A	0.008	N/A	N/A	N/A
27	0.019	N/A	N/A	0.021	N/A	N/A	N/A
28	0.004	N/A	N/A	0.007	N/A	N/A	N/A
29	0.022	N/A	N/A	0.025	N/A	N/A	N/A
30	0.004	N/A	N/A	0.007	N/A	N/A	N/A
31	0.017	N/A	N/A	0.020	N/A	N/A	N/A
32	0.004	N/A	N/A	0.007	N/A	N/A	N/A
33	0.019	N/A	N/A	0.020	N/A	N/A	N/A
34	0.002	N/A	N/A	0.004	N/A	N/A	N/A
35	0.018	N/A	N/A	0.020	N/A	N/A	N/A
36	0.003	N/A	N/A	0.004	N/A	N/A	N/A
37	0.018	N/A	N/A	0.020	N/A	N/A	N/A
38	0.003	N/A	N/A	0.004	N/A	N/A	N/A
39	0.016	N/A	N/A	0.018	N/A	N/A	N/A
40	0.004	N/A	N/A	0.007	N/A	N/A	N/A

Note: Measured reference fundamental current limits were applied for this test.

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Schaffner Profline
San Diego, California

Voltage Source Verification Data (Run time)

EUT: Solar Inverter **Tested by: Puck**
Test category: Stage:2 1ph & NB 3ph(Rsce=33) **Test Margin: 100**
Test date: 2013-7-2
Test duration (min): 5
Comment: M/N:Omniksol-5k-TL2 Test Mode:33% Load
Customer: Omnik New Energy Co.ltd

Test Result: Pass **Source qualification: Normal**
Measured source distortion is within the requirements of the standards
Measurements are compliant with IEC/EN61000-3-12(2004:11) & IEC/EN61000-4-7

Highest parameter values during test:

Voltage (Vrms):	230.11	Frequency(Hz):	50.00
I_Peak (Amps):	9.302	I_RMS (Amps):	6.015
I_1-Ref (Amps):	5.843	Crest Factor:	1.552
Power (Watts):	-1374	Power Factor:	-0.994

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.096	0.920	10.40	OK
3	0.189	2.876	6.56	OK
4	0.042	0.920	4.54	OK
5	0.141	3.451	4.07	OK
6	0.029	0.920	3.17	OK
7	0.185	2.876	6.43	OK
8	0.033	0.920	3.59	OK
9	0.192	1.380	13.88	OK
10	0.029	0.920	3.20	OK
11	0.187	1.611	11.61	OK
12	0.037	0.690	5.40	OK
13	0.170	1.380	12.32	OK
14	0.032	0.690	4.64	OK
15	0.159	0.690	23.04	OK
16	0.033	0.690	4.76	OK
17	0.155	0.690	22.51	OK
18	0.021	0.690	3.08	OK
19	0.137	0.690	19.90	OK
20	0.021	0.690	3.08	OK
21	0.133	0.690	19.30	OK
22	0.027	0.690	3.94	OK
23	0.128	0.690	18.60	OK
24	0.033	0.690	4.83	OK
25	0.127	0.690	18.36	OK
26	0.039	0.690	5.66	OK
27	0.128	0.690	18.55	OK
28	0.037	0.690	5.31	OK
29	0.125	0.690	18.16	OK
30	0.034	0.690	4.93	OK
31	0.108	0.690	15.71	OK
32	0.044	0.690	6.34	OK
33	0.115	0.690	16.61	OK
34	0.027	0.690	3.92	OK
35	0.104	0.690	15.03	OK
36	0.017	0.690	2.53	OK
37	0.088	0.690	12.75	OK
38	0.016	0.690	2.26	OK
39	0.075	0.690	10.85	OK
40	0.013	0.690	1.91	OK

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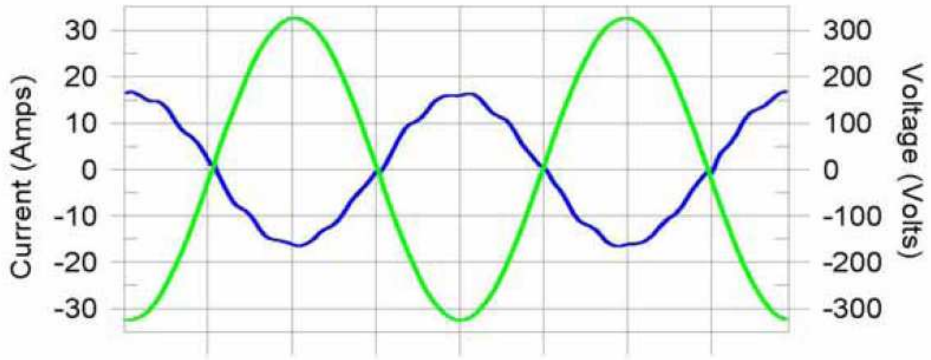
Schaffner Profline
San Diego, California

Harmonics – Per EN/IEC61000-3-12(Run time)

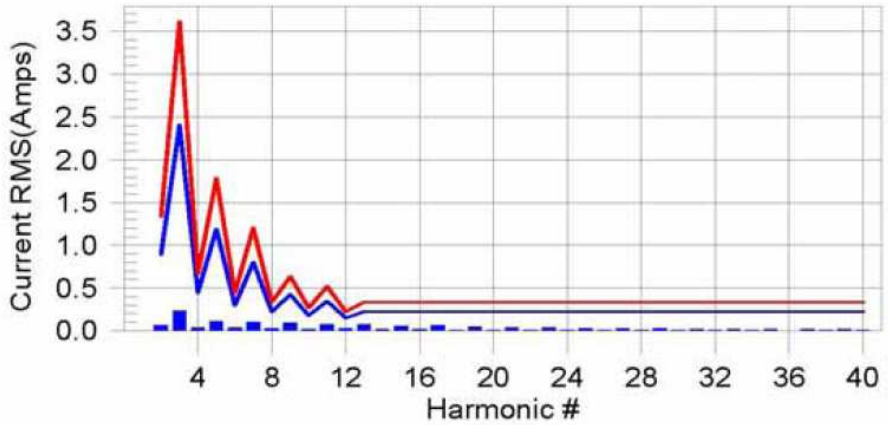
EUT: Solar Inverter **Tested by: Puck**
Test category: Stage:2 1ph & NB 3ph(Rsce=33) **Test Margin: 100**
Test date: 2013-7-2
Test duration (min): 5
Comment: M/N:Omniksol-5k-TL2 Test Mode:66% Load
Customer: Omnik New Energy Co.ltd

Test Result: Pass **Source qualification: Normal**

Current & voltage waveforms



Harmonics and Class 2 limit line **European Limits**



Test result: Pass **Worst harmonic was #13 with 30.30 % of the limit.**

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Annex 2

Pictures of the unit

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Omniksol-5k-TL2 - front view

Omniksol-5k-TL2 - side view

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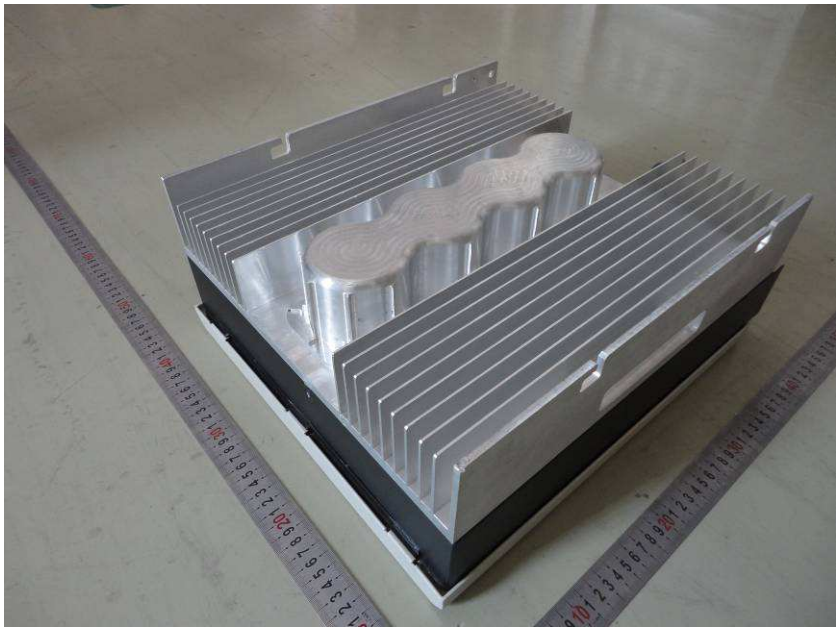
Omniksol-5k-TL2 rear



Omniksol-5k-TL2 - bottom view

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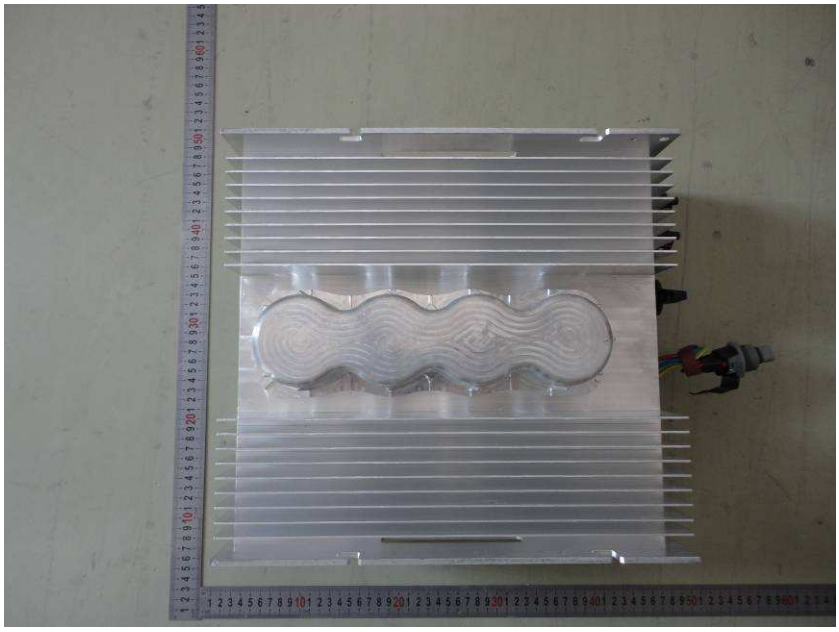
Omniksol-5k-TL2 - rear view



Omniksol-5k-TL2 - open view

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Photo for earthing connection

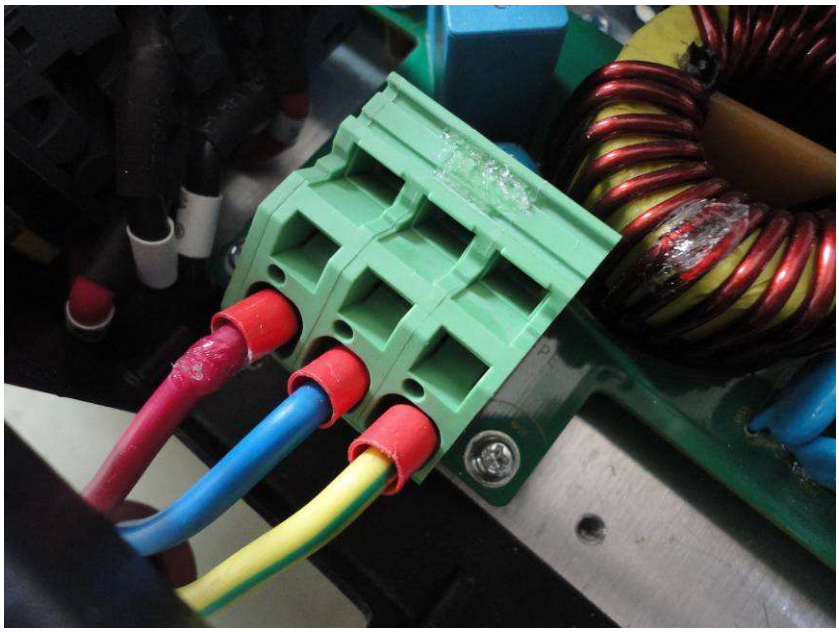

Photo of internal integrated RCD

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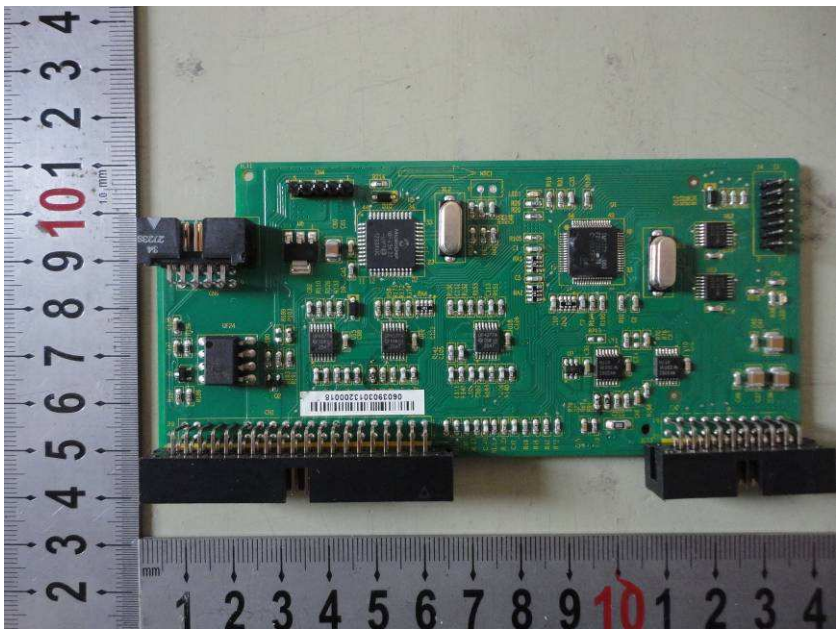
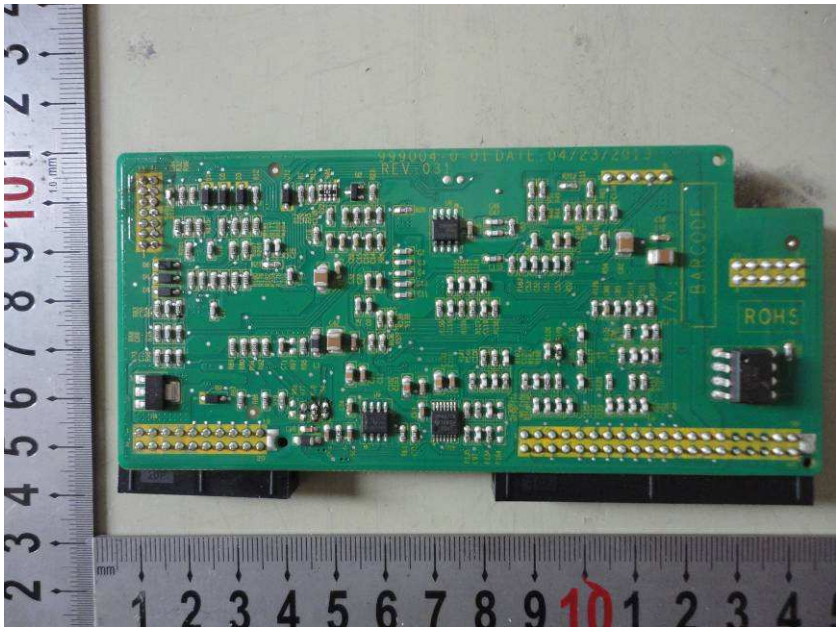
Omniksol-5k-TL2 - control board - component side



Omniksol-5k-TL2 - control board - Solder side



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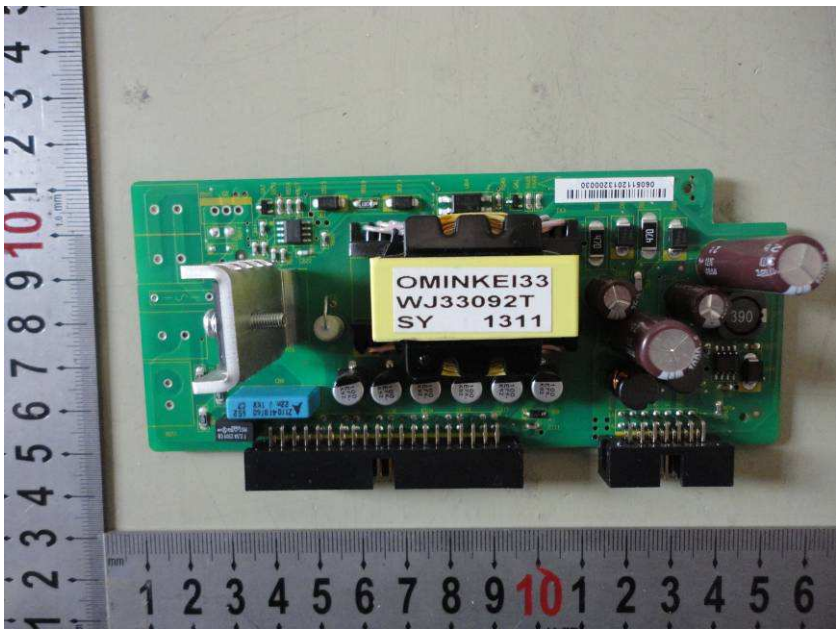
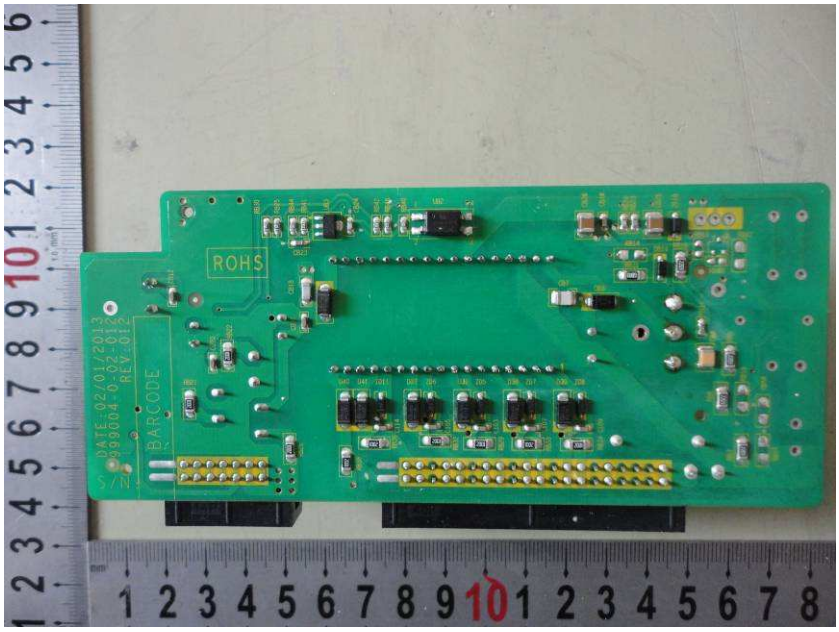
Omniksol-5k-TL2 - power supply board - component side



Omniksol-5k-TL2 - power supply board - Solder side



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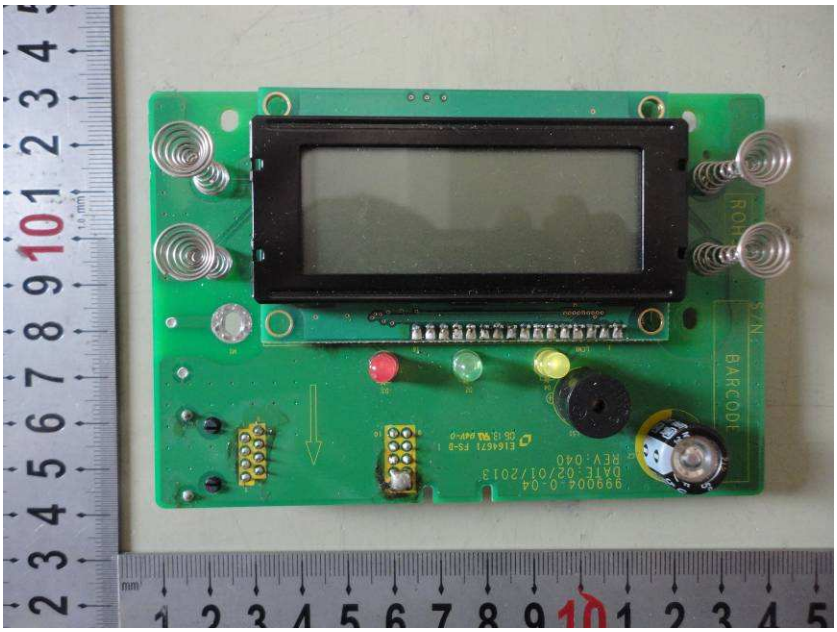
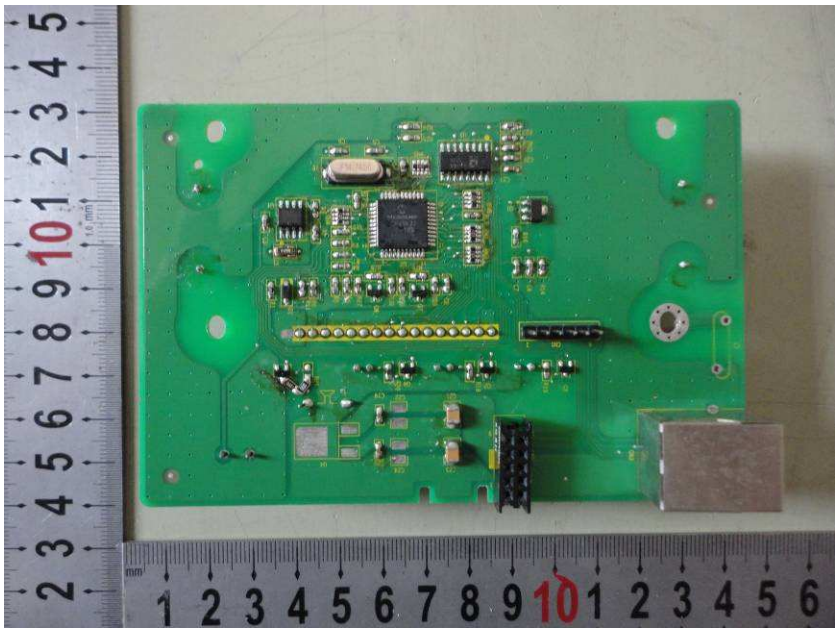
Omniksol-5k-TL2 - Display board - Component side



Omniksol-5k-TL2 - Display board - Solder side



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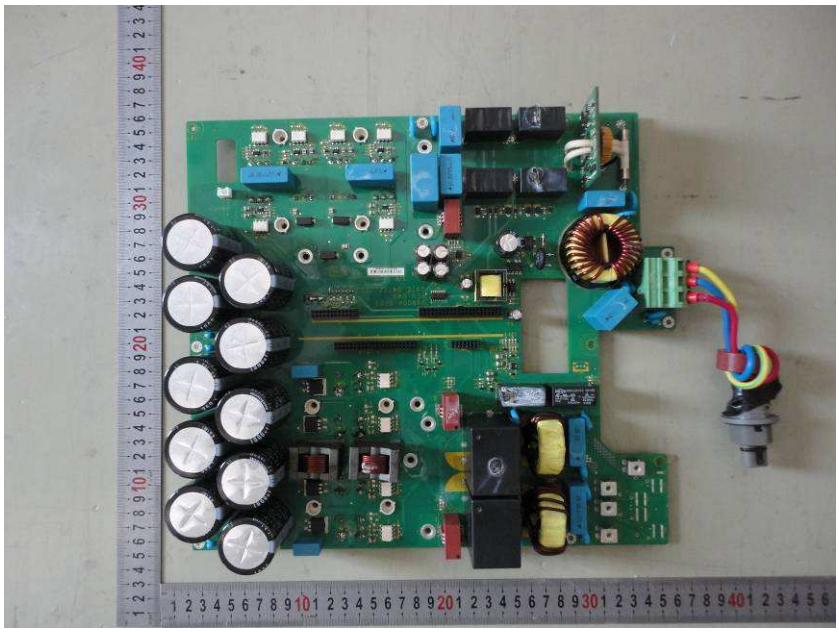
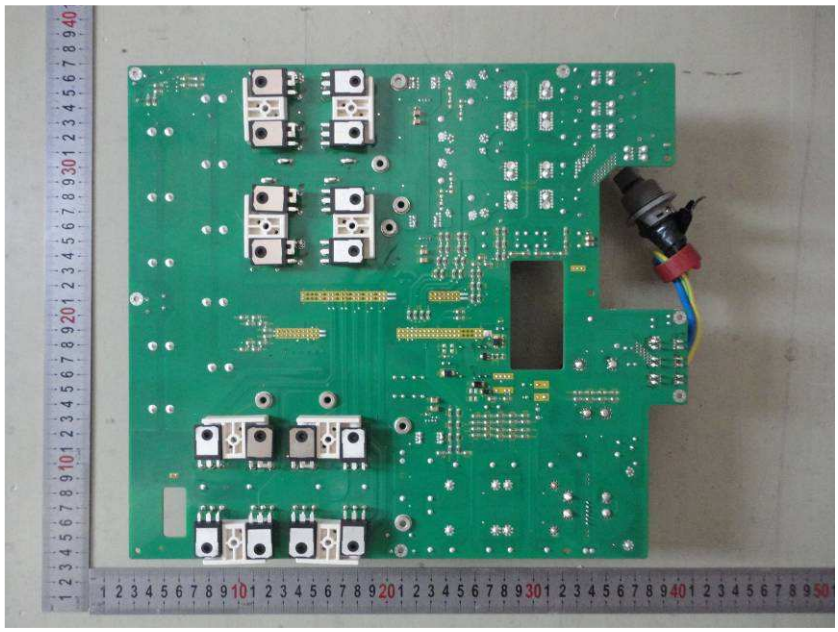
Omniksol-5k-TL2 - Main board - Component side



Omniksol-5k-TL2 - Main board - Solder side



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Annex 3

Test equipment list

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No.	Equipment	Internal No.	Type/characteristics	Manufacturer	Last Calibration	Due Data
1	Oscilloscope	A4089008SH	DPO3014	Tektronix	26-Mar-15	25-Mar-16
2	Oscilloscope	A4089024SH	P4034B	Tektronix	24-Sep-14	23-Sep-15
3	Oscilloscope	A4089035SH	TDS 2014C	Tektronix	04-Dec-14	03-Dec-15
4	High Voltage probe	A4089004SH	P2220	Tektronix	04-Dec-14	03-Dec-15
5	High Voltage probe	A4089009SH	P6139B	Tektronix	26-Mar-15	25-Mar-16
6	High Voltage probe	A4089044SH	P6015	Tektronix	02-Feb-15	01-Fec-16
7	High Voltage probe	A4089004SH	P2220	Tektronix	04-Dec-14	03-Dec-15
8	Current probe	A4089021SH	IT 700-S	LEM	14-Oct-14	13-Oct-15
9	Current probe	A4089017SH	TCP0150	Tektronix	24-Sep-14	23-Sep-15
10	AC power supply	A7040049SH	AFC-31010T	APC	26-Apr-15	25-Apr-17
11	AC power supply	A7040071SH	61512	Chroma	13-Mar-14	12-Mar-16
12	Programmable DC source	A7040058SH	62150H-1000S	Chroma	-	-
13	Programmable DC source	A7040059SH	62150H-1000S	Chroma	-	-
14	Programmable DC source	A7040069SH	62150H-1000S	Chroma	-	-
15	Programmable DC source	A7040070SH	62150H-1000S	Chroma	-	-
16	anti-isolating test stystem	A7150074SH	ACTL-380SH	qunling	-	-
17	Power Analyzer	A1240096SH	WT3000	YOKOGAWA	05-Feb-15	04-Feb-16
18	roll ruler	B1040160SH	5m	TAJIMA	28-Sep-14	27-Sep-15
19	load cabinet	A7150082SH	WSTF-LDJ150K/04S	shanghai wen shun	-	-
20	load cabinet	A7150083SH	WSTF-LDJ60K/300	shanghai wen shun	-	-
21	load cabinet	A7150077SH	WSTF-RC25k/0.3D 0.001kVA-25kVA	shanghai wen shun	-	-

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